



Multi-MW Targets for Next-Generation Accelerators

Robert Zwaska,

The 9th International Particle Accelerator Conference (IPAC '18)

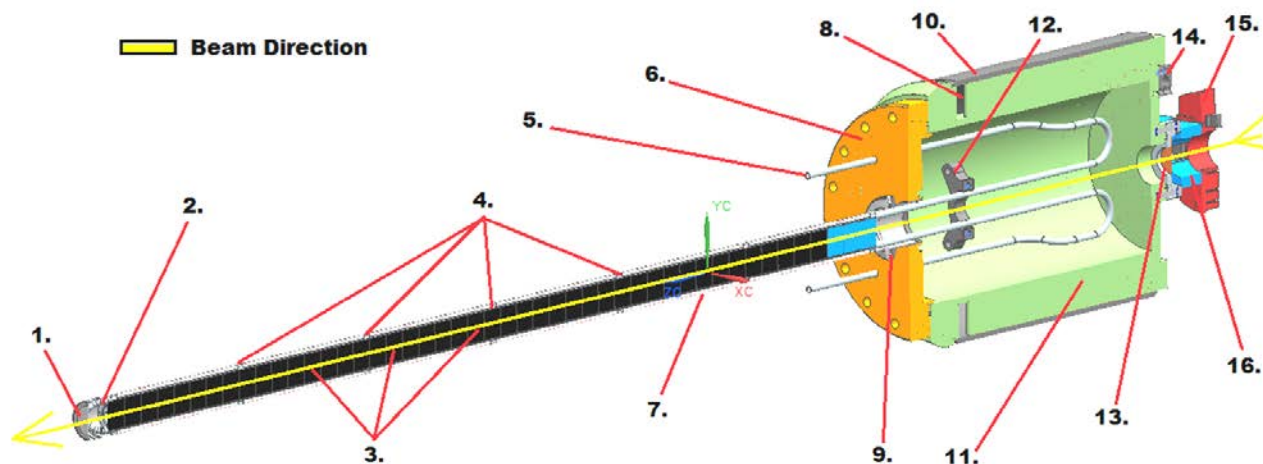
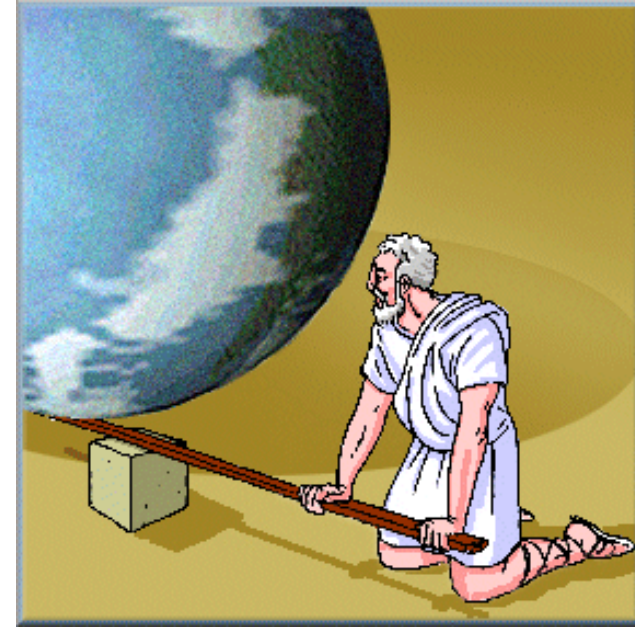
30 April 2018

Targets: The Fulcrum between Accelerator and Experiment

High-power particle beams are not generally interesting in themselves – they must be transformed into the particle of interest.

A **target** converts high-power beams into the particles of interest, whether they be pions, neutrinos, muons, neutrons...

Targetry includes the target, and all the other necessary optics, instrumentation, and infrastructure that are involved in the conversion.

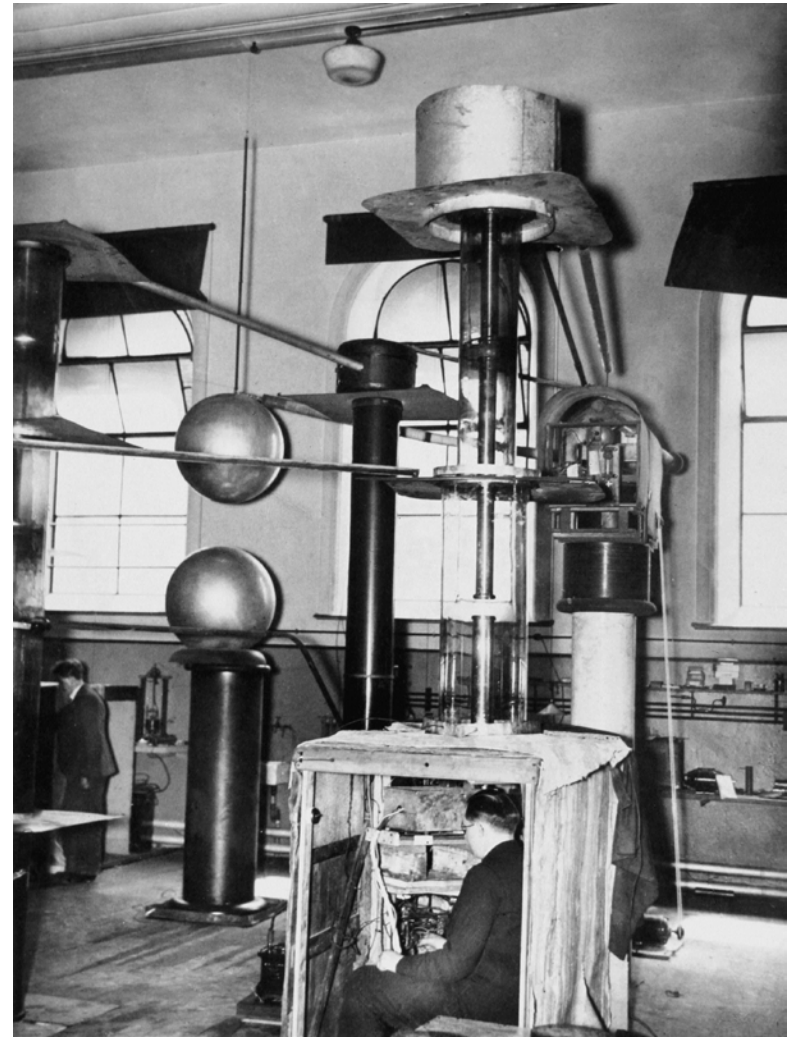
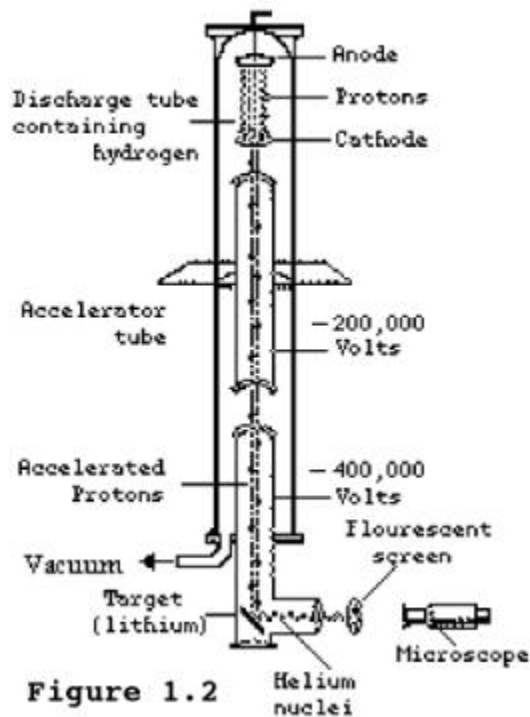


Outline & Summary

- Targets have a long history, but are entering a new regime with multi-MW beams (mostly protons)
- Numerous challenges will confront the efficient use of high-power beams
- Generic R&D can make an impact
 - Understanding of materials
 - Development of technologies
 - RaDIATE Collaboration formed to coordinate research
- Facilities world-wide are under active development and performing specific R&D and design

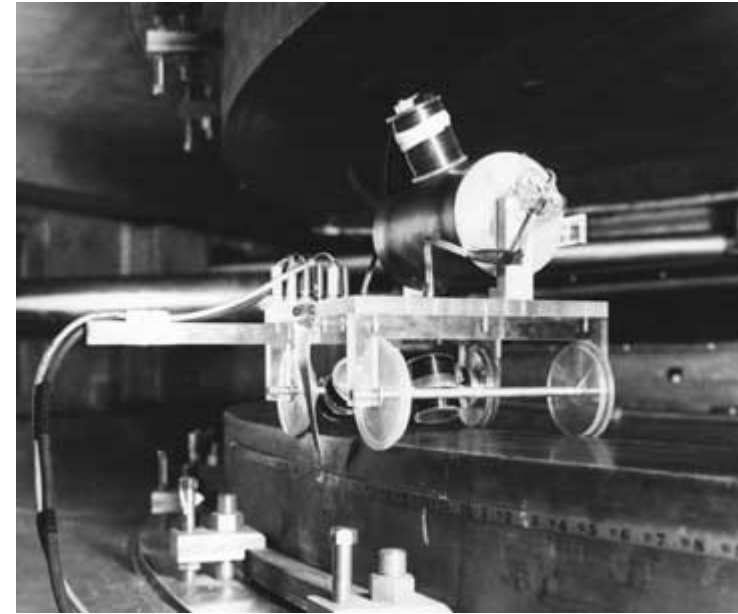
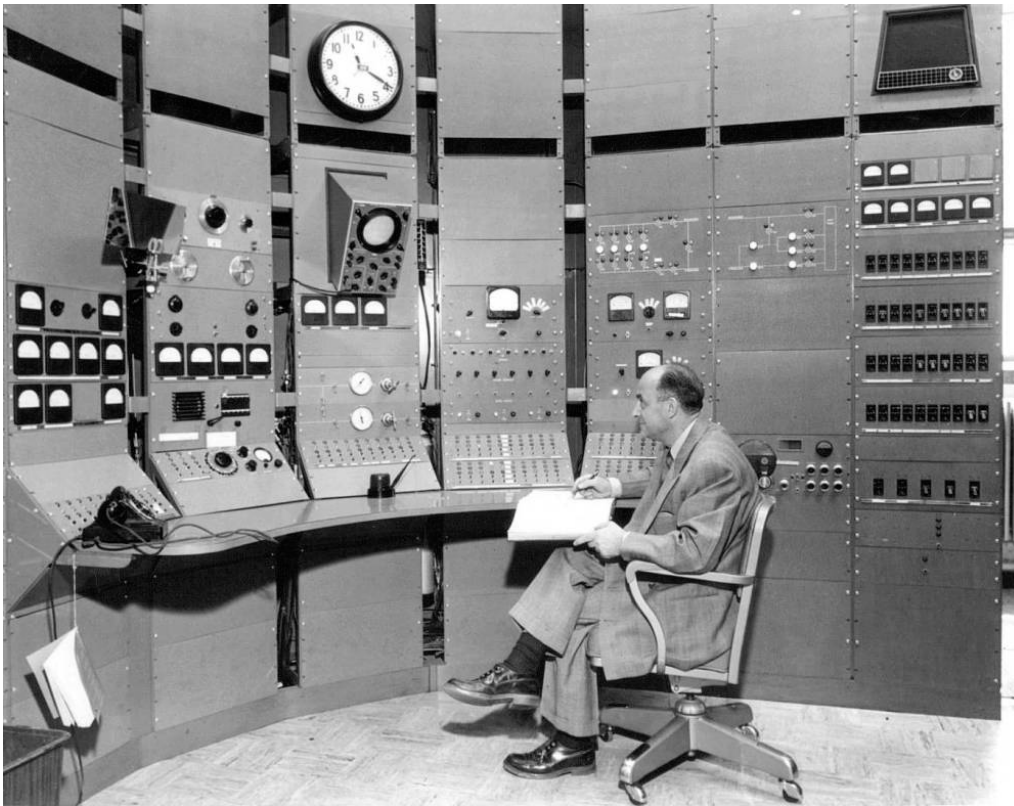
Targets have been in use since the start of accelerators

Lithium target used by Cockroft-Waltons to demonstrate atomic disintegration and neutron production



Enrico Fermi was an Early Targeteer

Designed and built his own remotely operated target trolley for the Chicago Syncrocyclotron

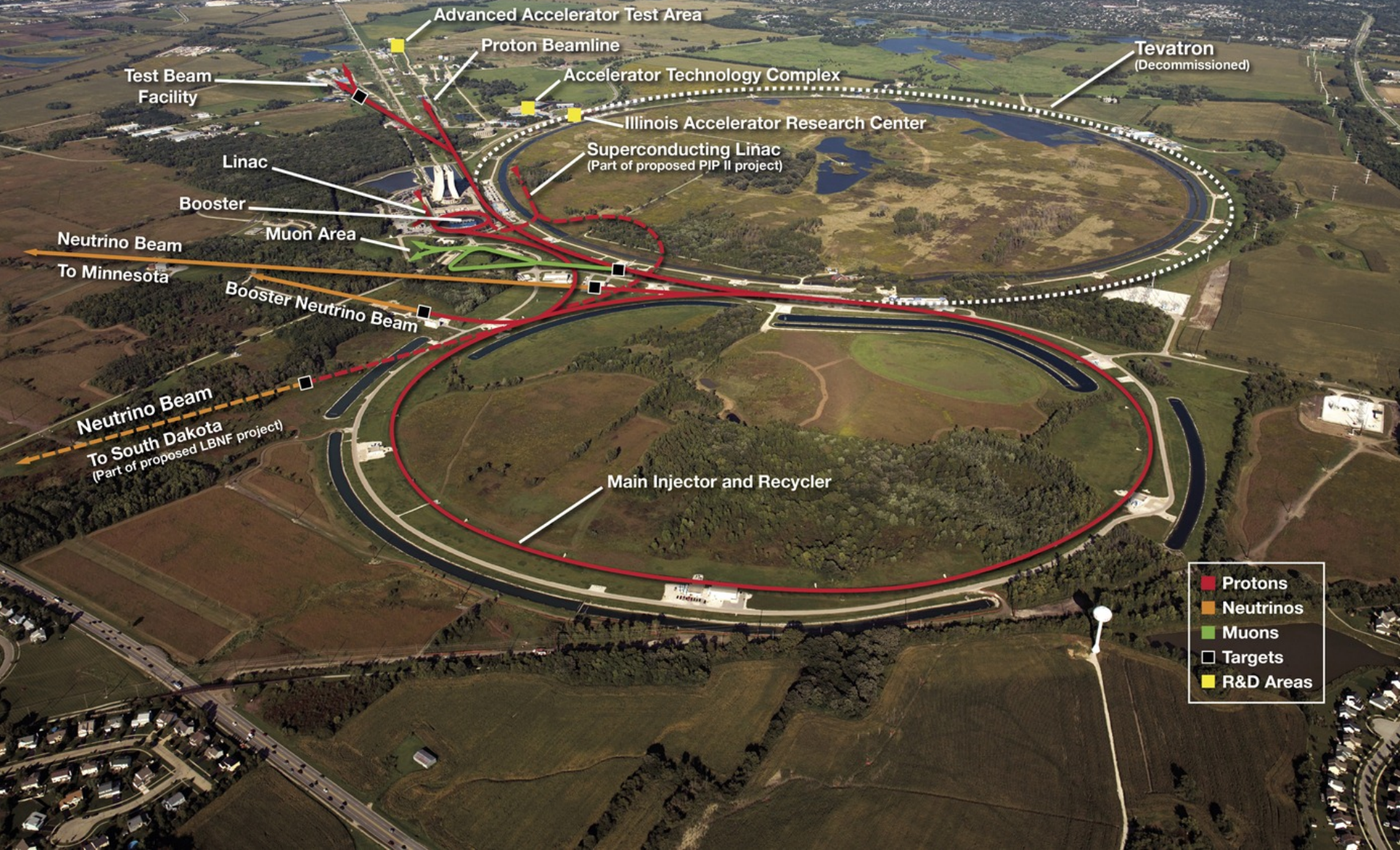


Often enjoyed operating the machine and trolley himself

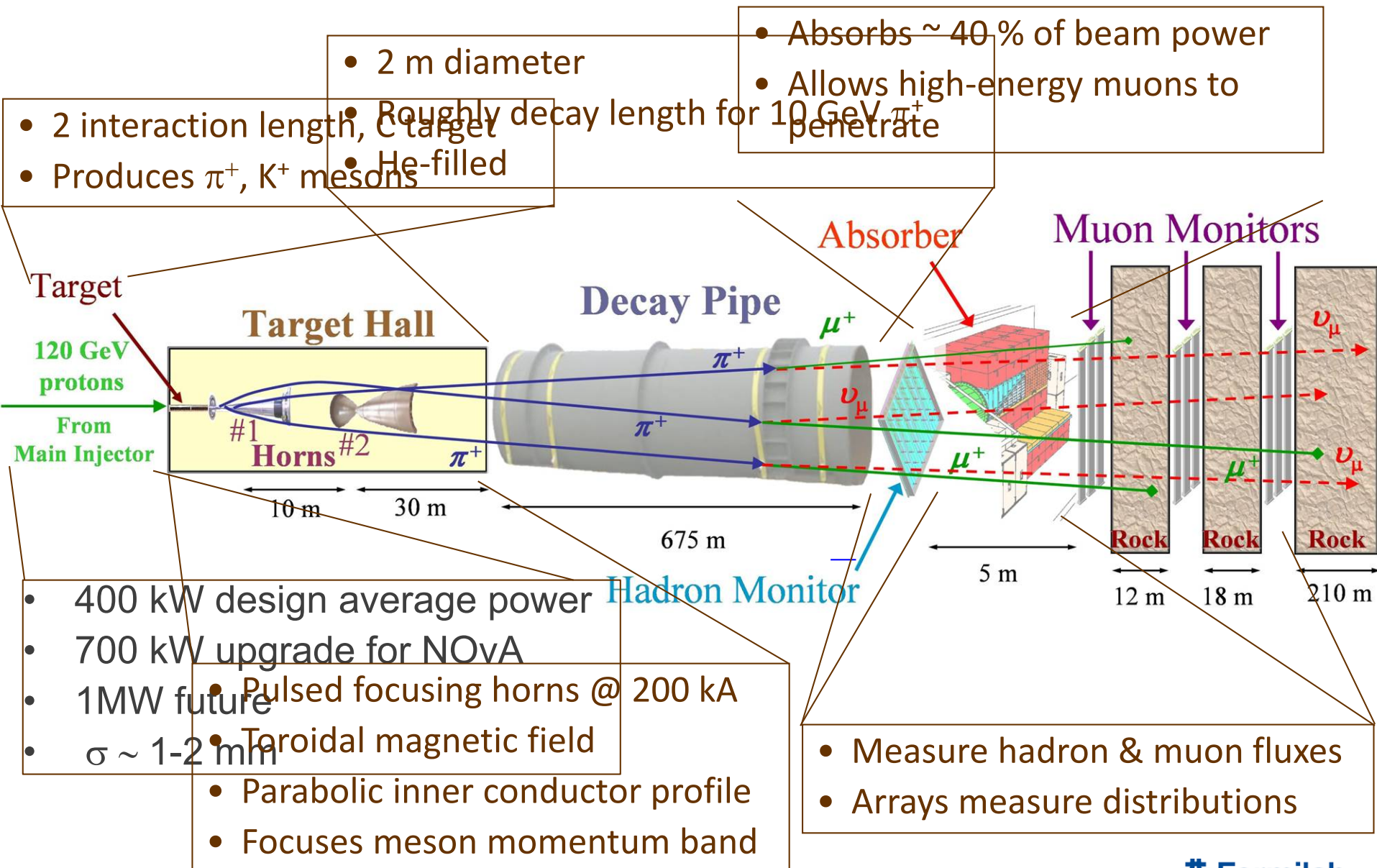
The Very First Target? – An ancient technology



Fermilab Accelerator Complex



Neutrino Beam in a Nutshell (NuMI)



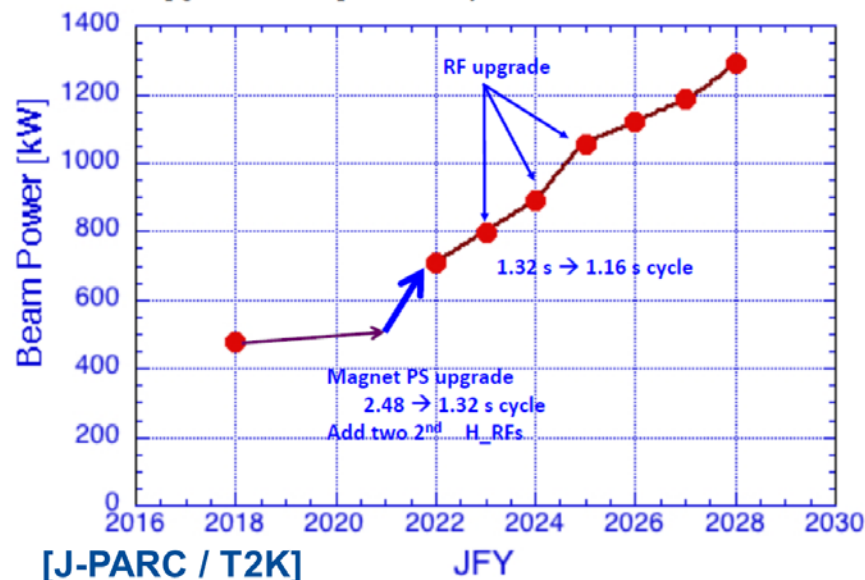
Multi-MW beams are coming

- Neutron Sources
- Neutrino Sources
- Fusion Material Irradiation
- Rare Ion Beams
- Accelerator Driven Systems
- ...

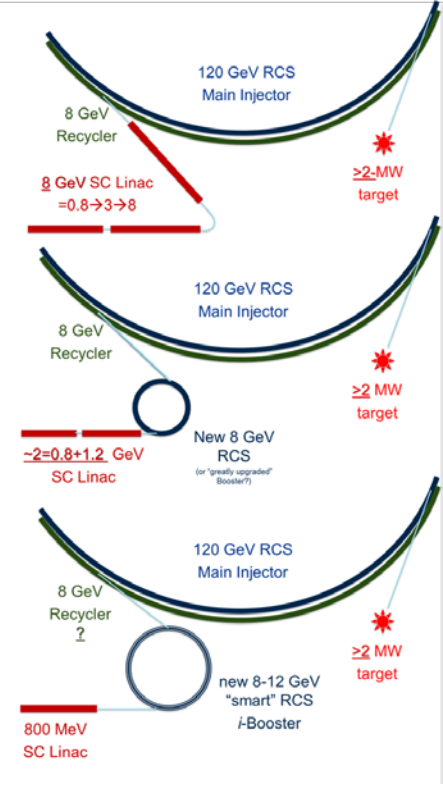
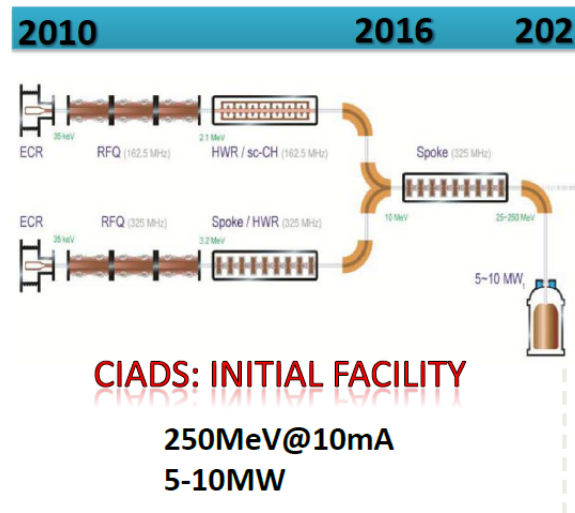
Mostly Proton beams in this talk

Projection for the Beam Power of 1.3 MW

- Major hardware upgrade is necessary for the new target of 1.3 MW.
- Hardware upgrade will be presented by Yoshii san.



Chinese ADS roadmap



High Power/Intensity Targetry Challenges

- Material Behavior
 - **Thermal “shock” response**
 - **Radiation damage**
 - Highly non-linear thermo-mechanical simulation
- Targetry Technologies (System Behavior)
 - Target system simulation (optimize for physics & longevity)
 - Rapid heat removal
 - Radiation protection
 - Remote handling
 - Radiation accelerated corrosion
 - Manufacturing technologies

HPT R&D Roadmap Workshop held at Fermilab to develop best R&D routes for next generation targetry facilities

- HPT R&D Roadmap workshop held in late May 2017 to come to consensus among the HPT community on best path forward
- Next generation target facility requirements were collected
- Most promising R&D cycles were identified
- Enabling R&D activities identified
- Workshop report available on web-site

http://indico.fnal.gov/e/HPT_Roadmap_Workshop_2017



Workshop Charge: *Identify the required and most beneficial High Power Targetry (HPT) R&D routes and associated key milestones, considering the needs, objectives, and timeline of the HEP experimental program as recommended in the 2014 P5 report.*



Radiation Damage In Accelerator Target Environments

Broad aims are threefold:

radiate.fnal.gov

- to generate new and useful materials data for application within the **accelerator** and **fission/fusion** communities
- to recruit and develop new scientific and engineering experts who can **cross the boundaries** between these communities
- to initiate and coordinate a **continuing synergy** between research in these communities, benefitting both **proton accelerator applications** in science and industry and **carbon-free energy technologies**



Roadmap Workshop Recommendation for Prioritization

Prioritization should be given to fundamental **Material R&D over Technological R&D** in cases of limited resources

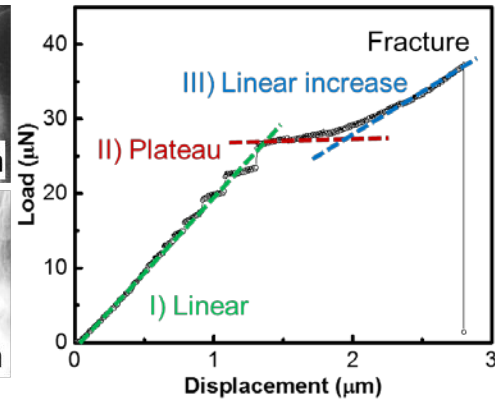
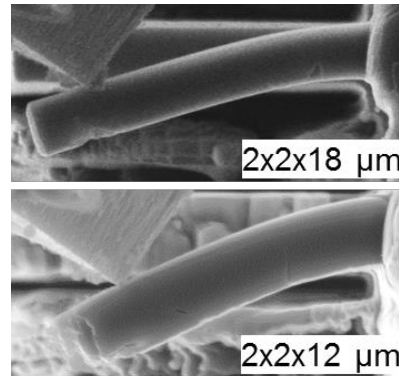
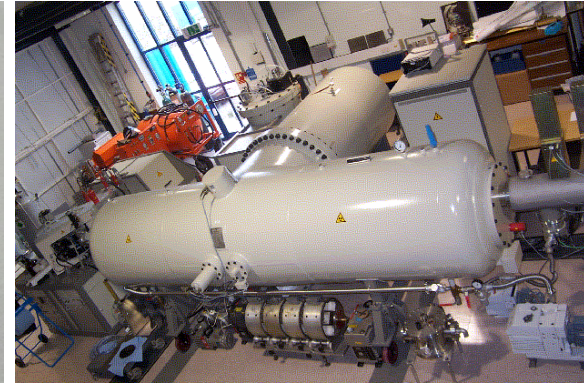
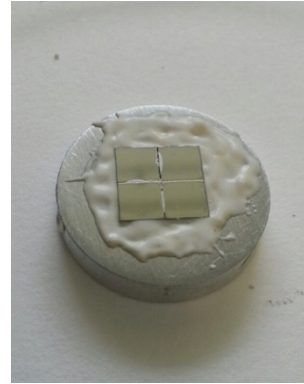
- **Materials R&D** topics (e.g. radiation damage effects and thermal shock response) have higher priority due to:
 - the general lack of a knowledge base for these applications;
 - the relatively long time frame for R&D iteration cycles;
 - and because individual projects cannot support such long-term, fundamental research studies.
- **Targetry Technology** studies (e.g. remote handling, radiation protection, quality assurance, manufacturing technologies) generally have lower priority due to:
 - existing knowledge base currently being advanced in industry;
 - typically shorter R&D iteration cycles;
 - and because individual projects must support these highly facility-specific development efforts to deliver a credible Conceptual Design Report to their sponsors.

Roadmap Workshop Material R&D Highlights

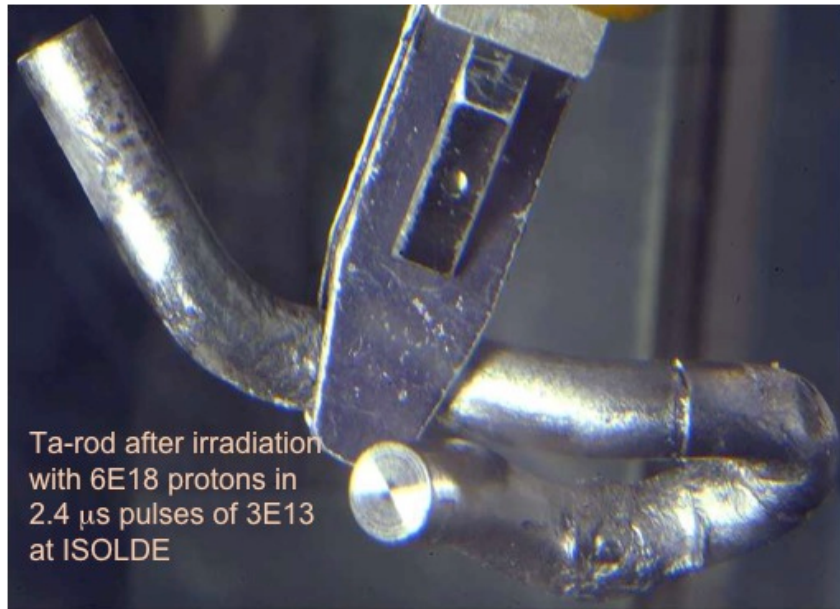
- Qualification of materials for use in HPT facilities will require **high-energy proton irradiations and thermal shock testing** in concert to reproduce single pulse material response (*HiRadMat BeGrid2 2018 experiment*)
- **High-cycle fatigue** response of materials in an **irradiated condition** is almost unique to the HPT accelerator facility application (*Fatigue specimens in BLIP 2017-18 irradiation*)
- **Autopsy and PIE of exposed target components** is a significant component of the radiation damage R&D thrust. New facilities should seriously consider including autopsy and irradiated material recovery capability during the design phase (*Autopsies of spent NuMI targets*)
- The current routes for **high-energy proton irradiations are expensive**, long in duration, and lack control of testing conditions and schedule. Alternative solutions/facilities should be explored. (*Opportunity for enabling R&D*)
- Likewise, **proton beam thermal shock studies are relatively expensive**, long in duration, and dependent upon the host accelerator facility operations schedule. Alternative solutions such as electron beam or laser beam facilities should be explored. (*Opportunity for enabling R&D*)

Roadmap Workshop Recommendations for Enabling R&D Activities

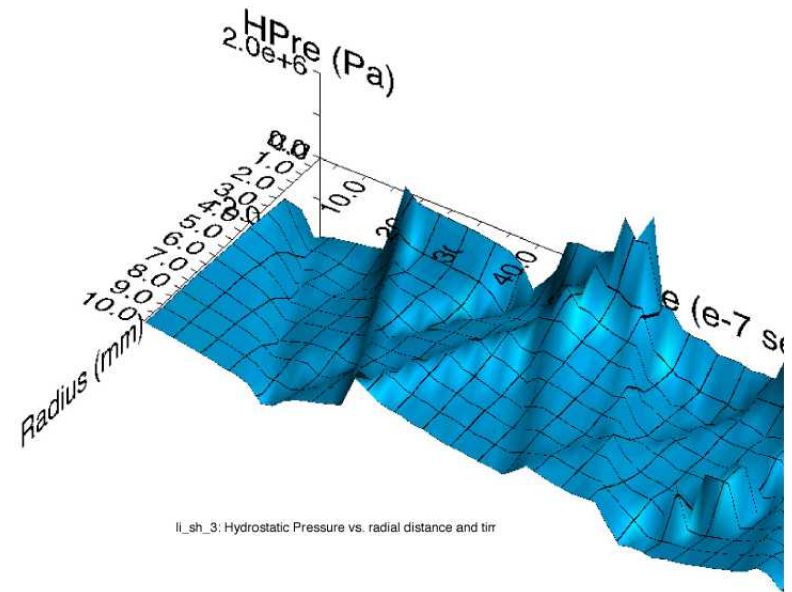
- **Low-energy ion irradiations** are attractive because they allow study of the evolution of the micro-structure during irradiation without activating the specimens, are relatively low cost, and can achieve high dose in very short durations.
- **Micro-mechanics** and meso-scale testing are potential enabling technologies to overcome some of the limitations of low-energy ion irradiations as well as to drastically reduce specimen size requirements (which also reduces activity of specimens).
- **Replicate thermal shock** loading from proton beam using electron beams, lasers, or other techniques could reduce the cost and length of R&D cycles compared to proton beam-line tests.



Material R&D: Thermal Shock (stress waves)

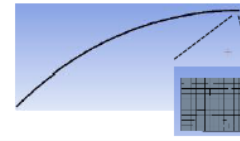
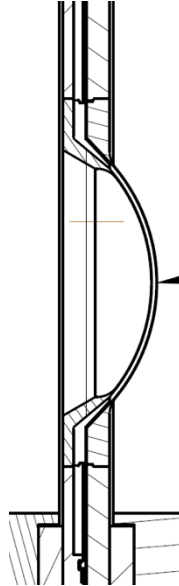


Ta-rod after irradiation with $6E18$ protons in $2.4 \mu s$ pulses of $3E13$ at ISOLDE (photo courtesy of J. Lettry)

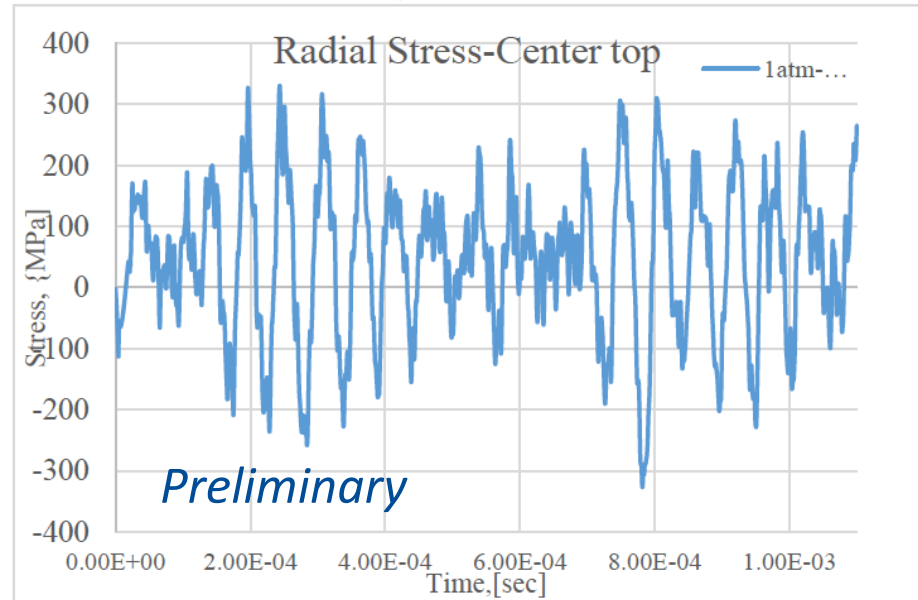


- Fast expansion of material surrounded by cooler material creates a sudden local area of compressive stress
- Stress waves (not shock waves) move through the target
- Plastic deformation, cracking, and fatigue can occur

Material R&D: Stress wave example: T2K window



1 atm. is applied on the concave side

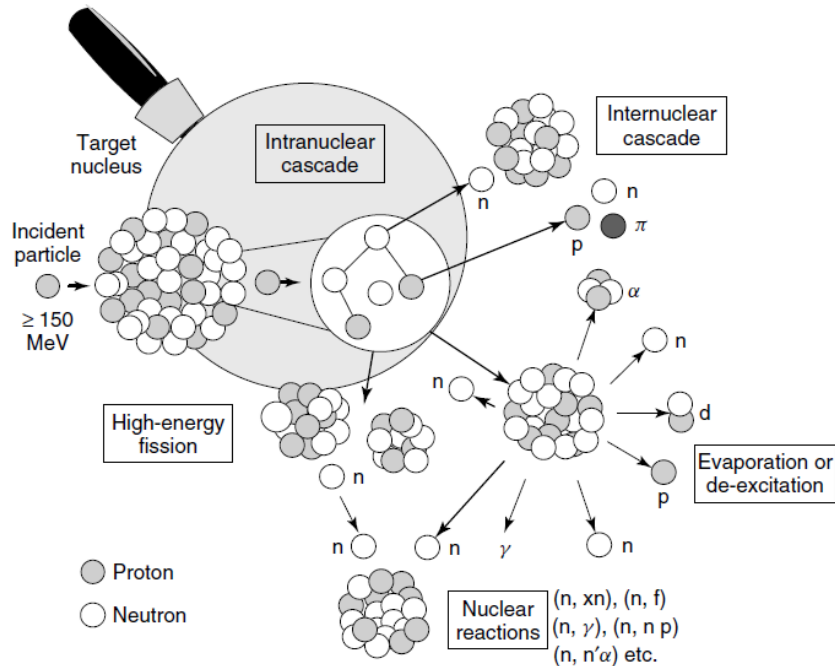


S. Bidhar, FNAL

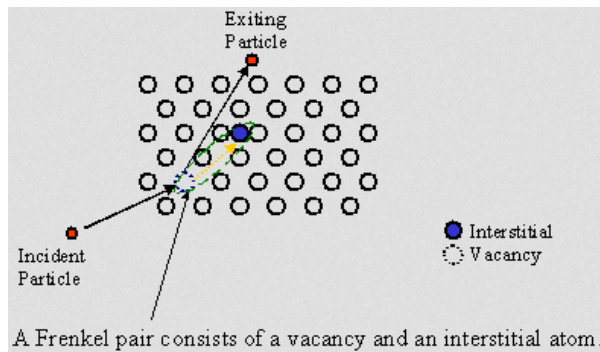
- Material response dependent upon:
 - Specific heat (temperature jump)
 - Coefficient of thermal expansion (induced strain)
 - Modulus of elasticity (associated stress)
 - Flow stress behavior (plastic deformation)
 - Strength limits (yield, fatigue, fracture toughness)

***Heavy dependence upon material properties, but:
Material properties dependent upon Radiation Damage...***

Material R&D: Radiation Damage Disorders Microstructure

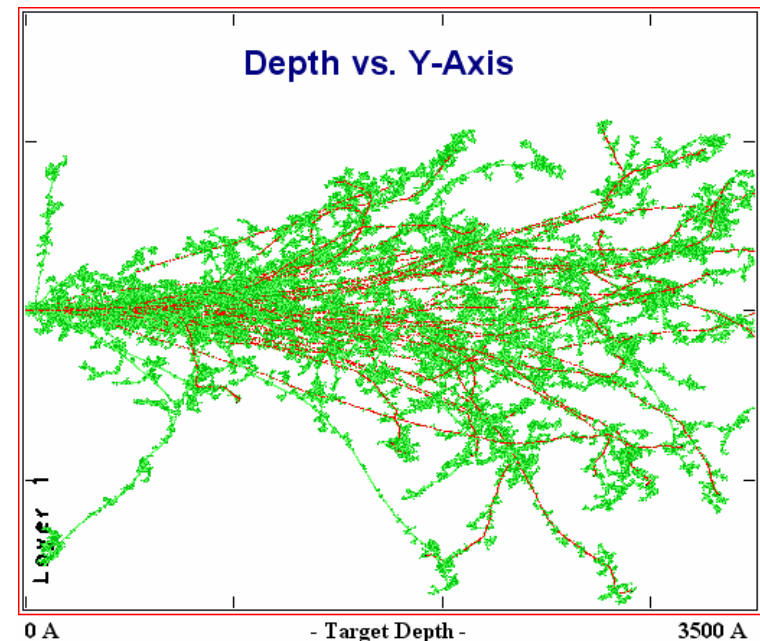


From D. Filges, F. Goldenbaum, in: *Handb. Spallation Res.*, Wiley-VCH Verlag GmbH & Co. KGaA, 2010, pp. 1–61.



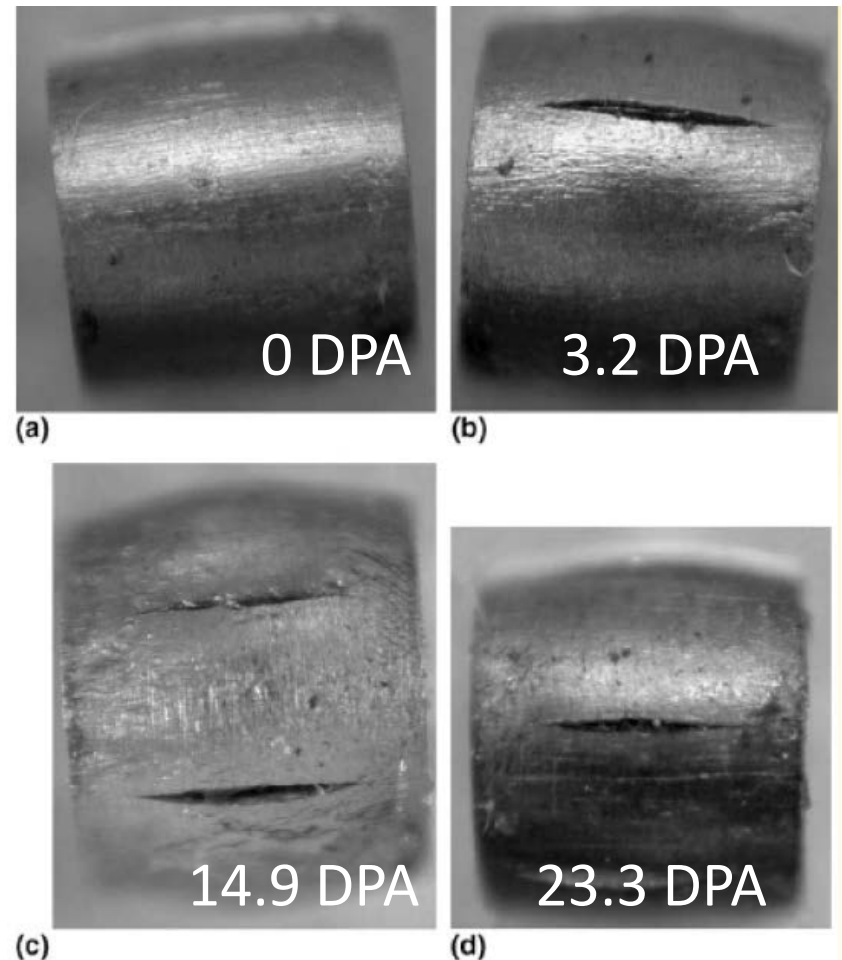
Microstructural response:

- *creation of transmutation products;*
- *atomic displacements (cascades)*
 - *average number of stable interstitial/vacancy pairs created = DPA (Displacements Per Atom)*



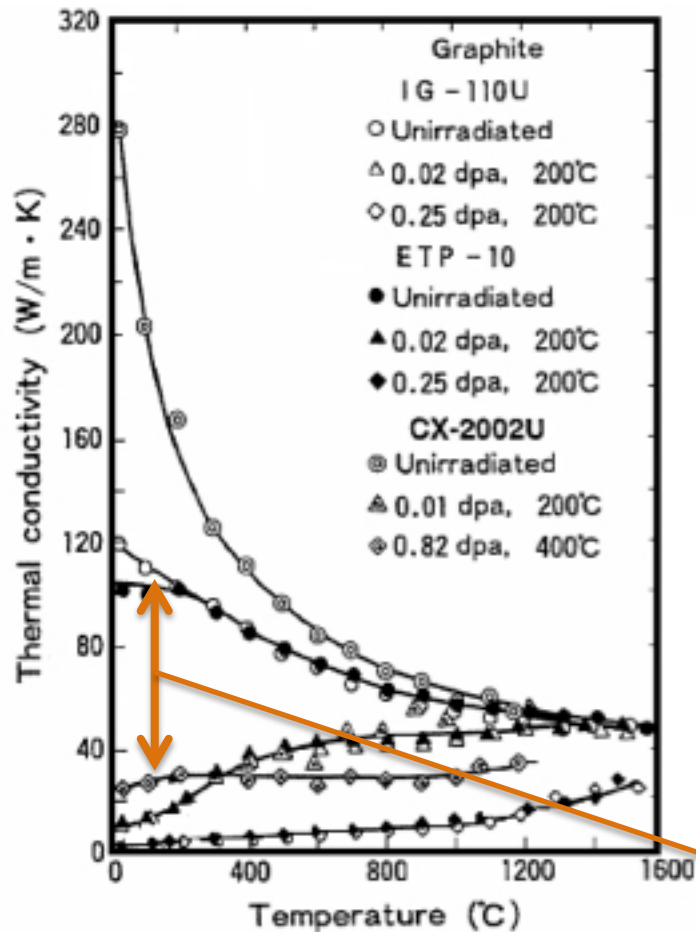
Material R&D: Radiation Damage Effects

- Displacements in crystal lattice
(expressed as Displacements Per Atom, DPA)
 - Embrittlement
 - Creep
 - Swelling
 - Fracture toughness reduction
 - Thermal/electrical conductivity reduction
 - Coefficient of thermal expansion
 - Modulus of Elasticity
 - Fatigue response
 - Accelerated corrosion
 - Transmutation products
 - H, He gas production can cause void formation and embrittlement
(expressed as atomic parts per million per DPA, appm/DPA)
- Very dependent upon material condition and irradiation conditions (e.g. temp, dose rate)



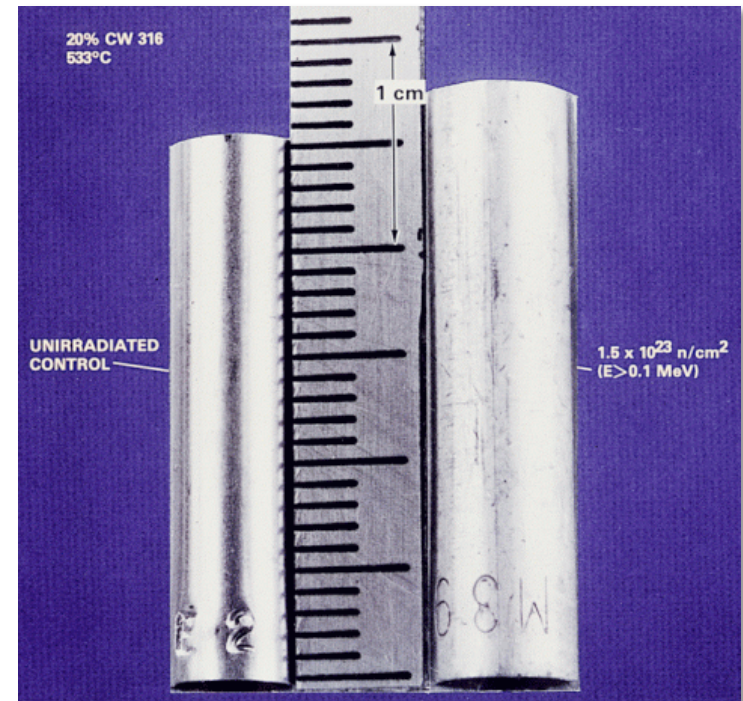
S. A. Malloy, et al., Journal of Nuclear Material, 2005. (LANSCE irradiations)

Material R&D: Radiation damage effects can be significant



N. Maruyama and M. Harayama, "Neutron irradiation effect on ... graphite materials," Journal of Nuclear Materials, 195, 44-50 (1992)

Factor of 10 reduction in conductivity at 0.02 DPA

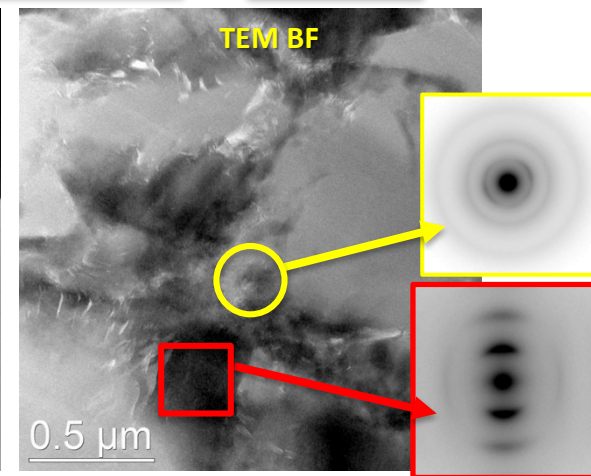
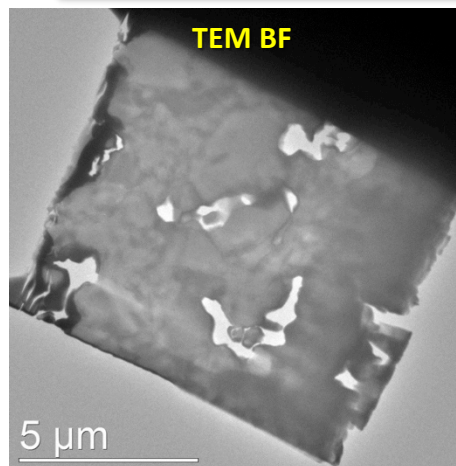


D.L. Porter and F. A. Garner, J. Nuclear Materials, 159, p. 114 (1988)

Void swelling in 316 Stainless steel tube (on right) exposed to reactor dose of 1.5E23 n/cm²

Autopsy of NuMI graphite target (Radiation Damage Separate Effects)

- NuMI target (NT-02) autopsy and graphite PIE (FNAL, PNNL)
 - Graphite fins saw $8E21$ p/cm² fluence
- Evidence of Bulk Swelling
 - The micrometer measurements indicate swelling did occur
 - More swelling in US fin locations
 - More swelling is associated with the fractured fins
- Evidence of fracture during operation
 - Symmetric fracture structure
 - Limited impurity transport into whole fins relative to fractured fins
- Evidence of limited radiation damage and material evolution
 - Surface discoloration appears to be mostly solder and flux material
 - Crystal structure & porosity consistent with non-irradiated state, perhaps explained by:
 - nano-crystalline features pinning defects
 - Extreme dose rate from pulsed beam



- Taken from fracture surface at the center where the beam was targeted
- Lamella has mixed regions of what appear to be amorphous (yellow insert diffraction pattern) and nanocrystalline microstructure (red square)
- Mrozowski cracks at the interfaces between these two regions

HE Proton Irradiations (Radiation Damage Separate Effects)

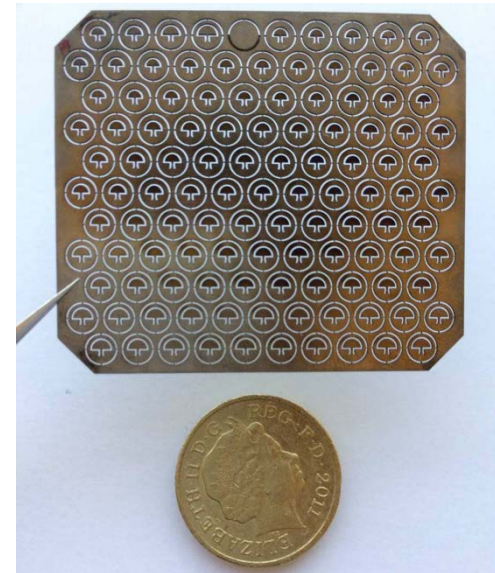
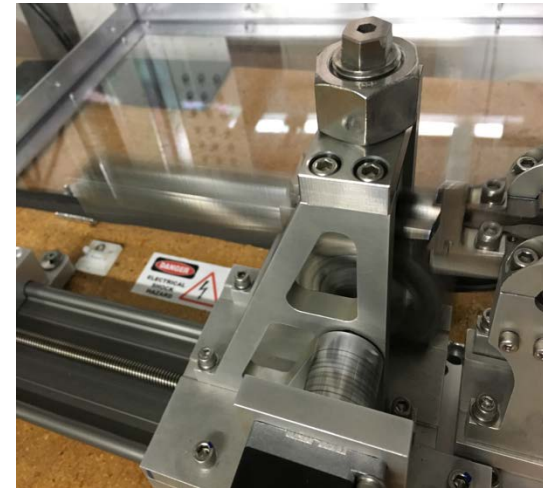
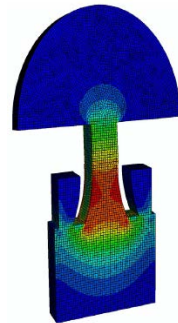
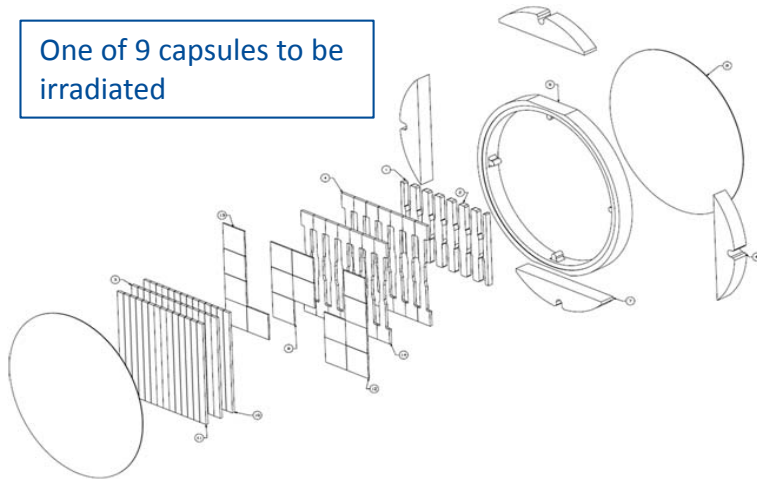
Future work includes 2017-18 BLIP irradiation

- Graphite at various temp (up to $\sim 1,000$ °C) to explore annealing of radiation damage
- Also Beryllium, Ti alloys, Si, TZM, Al, & Ir
- Post-Irradiation Examination (2018) includes mechanical, thermal, micro-structural, etc.
- Participants: FNAL, BNL, PNNL, FRIB, ESS, CERN, J-PARC, STFC, Oxford, LANL

Fatigue evaluation of irradiated Ti alloys (US-JP)

- World's first HE proton irradiated Ti high cycle fatigue study
- Miniature bending fatigue specimens (20 Hz)
- Meso-scale (few mm) fatigue specimens (20,000 Hz)

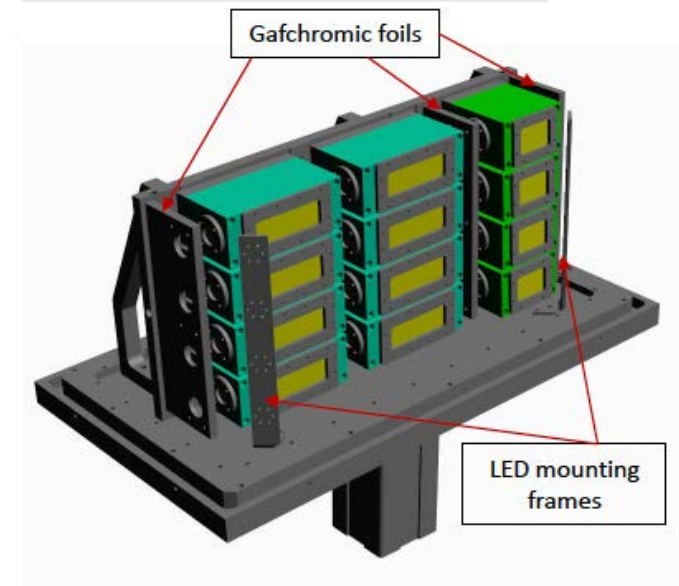
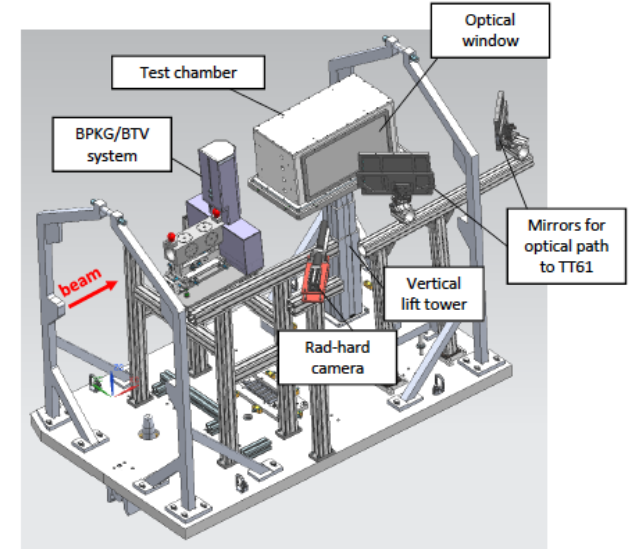
One of 9 capsules to be irradiated



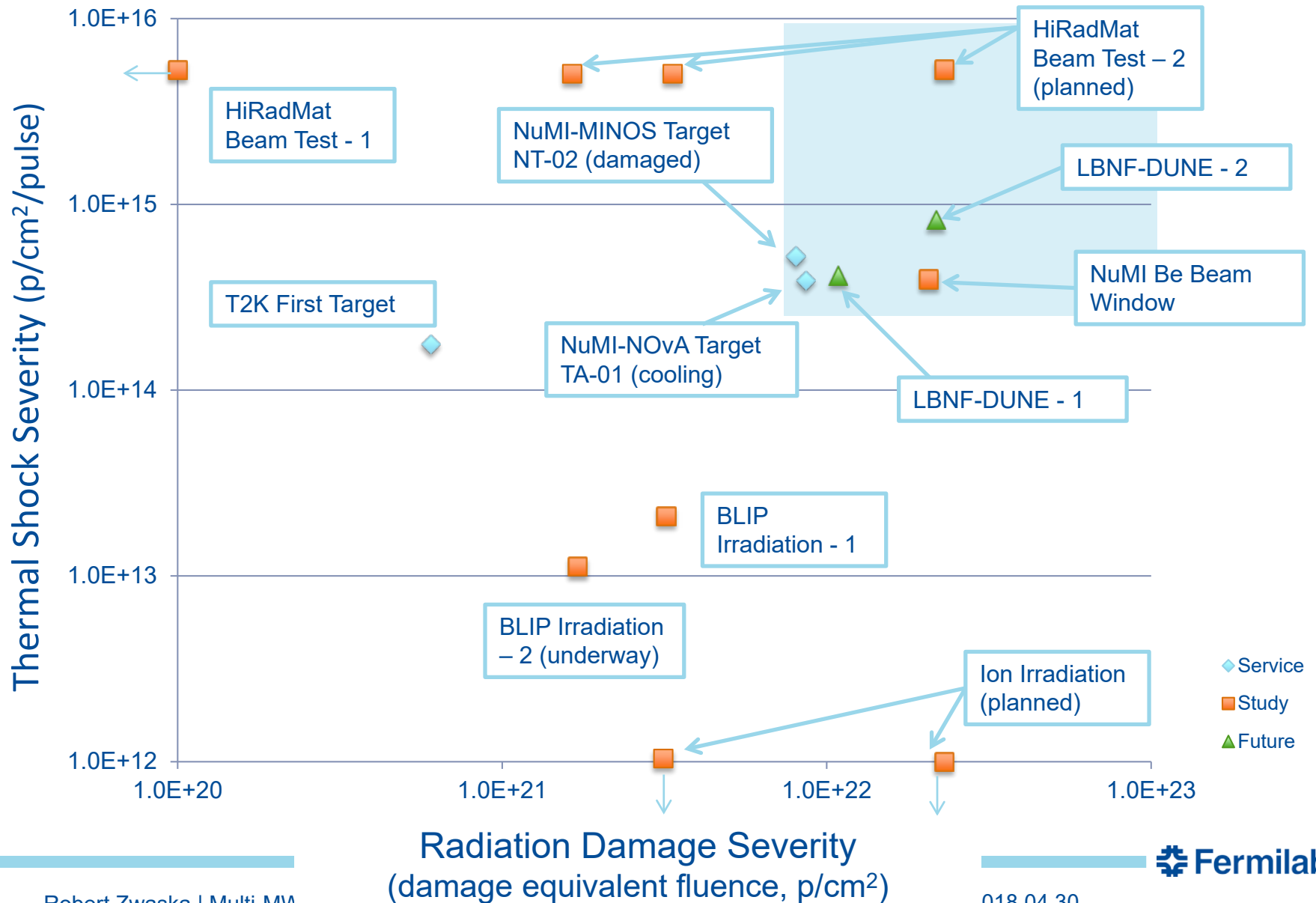
In-beam Thermal Shock of Irradiated Mat'ls (Combined effects)

BeGrid2 Experiment at HiRadMat (CERN):

- Testing of irradiated materials at BLIP
 - Beryllium grades
 - Graphite grades
 - Glassy Carbon
 - Silicon
- First time for irradiated materials at HiRadMat
- Higher p beam intensities than BeGrid1
- Non-irradiated novel materials (nano-fiber mats)
- Non-irradiated high-Z materials for CERN's Beam Dump Facility (SHiP)
- Experiment scheduled for October 2018

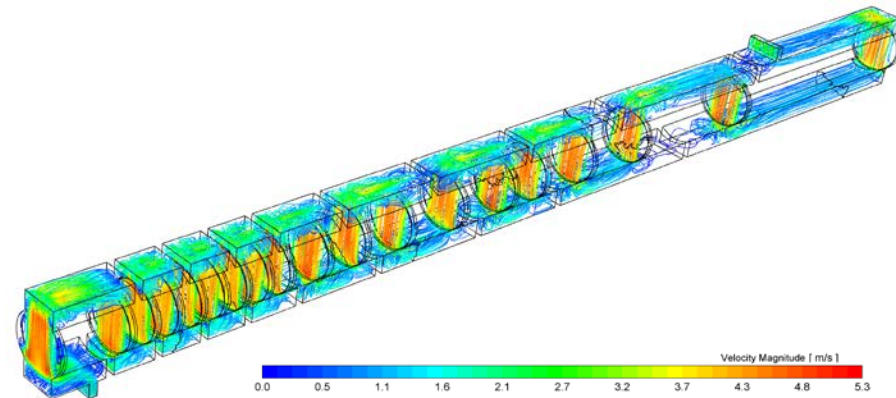
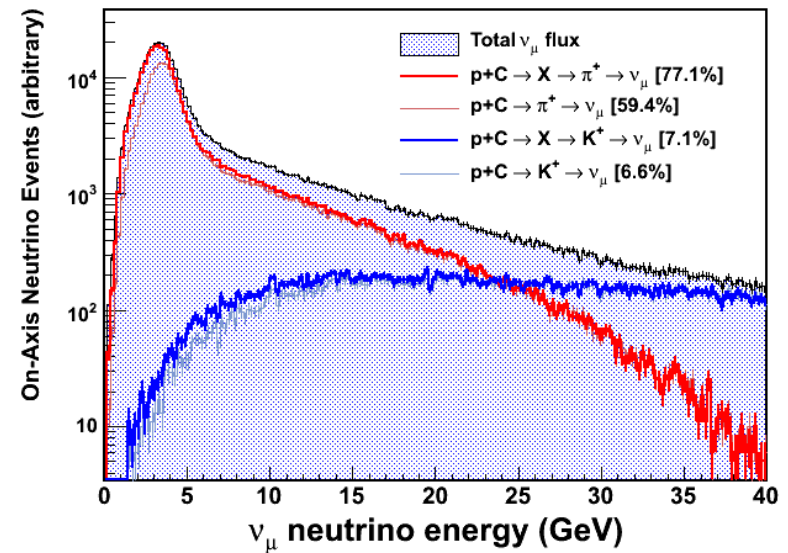


Nu HPT R&D Materials Exploratory Map



The Role(s) of Modeling and Simulation

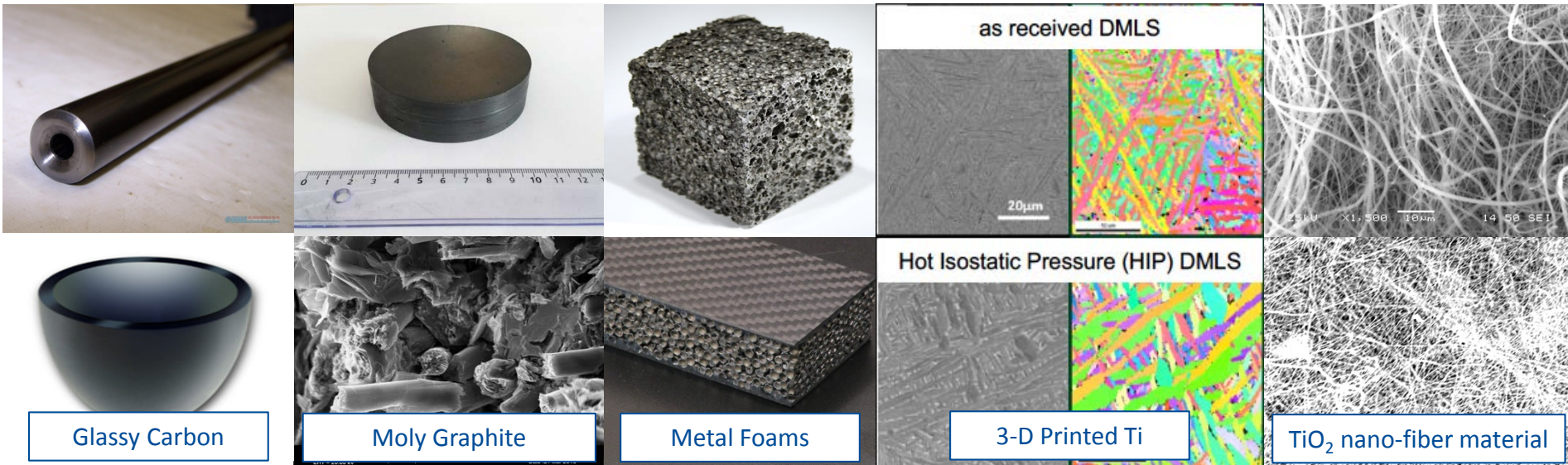
- MC codes used for energy deposition, radionuclide production, and DPA estimation (MARS, FLUKA, etc.)
 - Particle production has some key question
- Thermo-mechanical, dynamic modeling used for simulation of material behavior (ANSYS, LS-DYNA, Auto-Dyn, etc.)
- Empirical data from testing to provide radiation damaged properties of materials
- Molecular dynamics and crystal plasticity modeling is not yet mature enough to model atomistic changes, micro-structural evolution and subsequent effect on macro-properties (however the nuclear materials community is hard at work on these issues)
- Modeling of micro-structural features may be helpful to correlate micro-mechanics testing results with macro-properties and is also the subject of research in the nuclear materials community



Development of New Targetry Materials

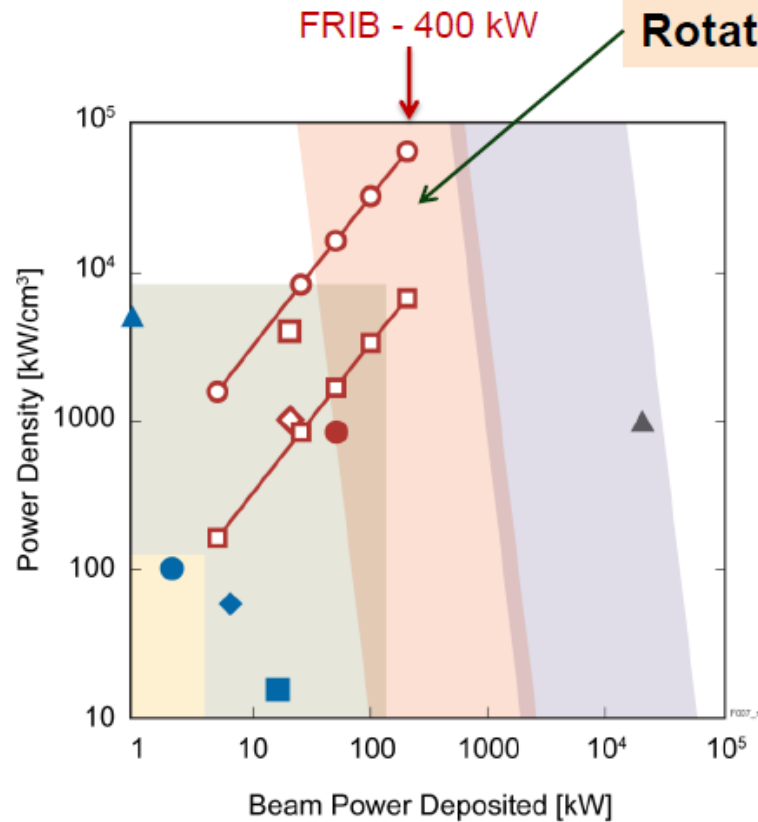
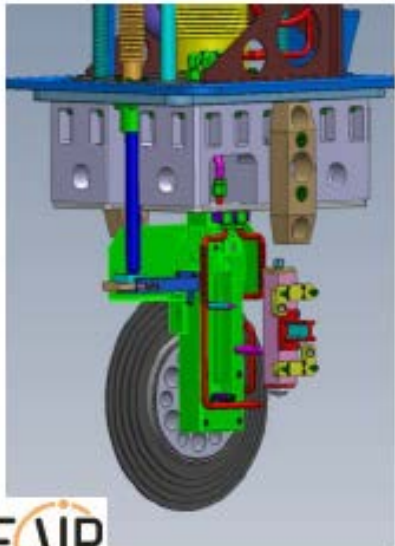
An ultimate objective is to develop new materials specifically addressing the requirements of future target facilities. Some progress is being made in exploring some of the newer materials and forms of material that have been developed.

- Glassy carbon (RaDIATE BLIP run material, CERN/FNAL)
- Molybdenum graphite (RaDIATE BLIP run material, CERN)
- Metal foams (BeGrid2 material, FNAL)
- 3-D Printed Ti alloy (RaDIATE BLIP run material, FRIB)
- Nano-fiber mats (BeGrid2 material and subject of Sujit Bidhar's talk, FNAL LDRD)



FRIB/FAIR building off cyclotron experience (PSI / TRIUMF)

- Moving from static, to rotating, to multiple-sliced targets
 - Potentially liquid in future



Running devices

- MSU - CCF
- ◆ SPIRAL - GANIL
- ▲ SISSI - GANIL
- LAMPF

Existing Prototypes

- ◇ GSI - FAIR
- RIKEN - RIBF
- FRIB - U
- ◻ FRIB - O
- SPIRAL2 - GANIL

To be developed

- ▲ IFMIF

EVA Phase – Target Facility



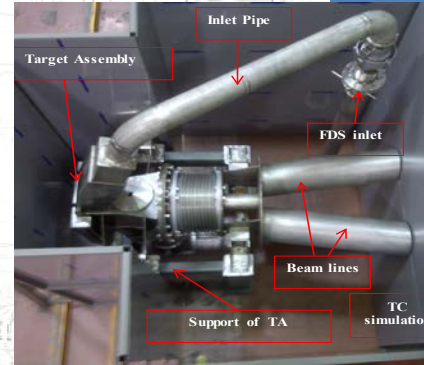
LIFUS6 in Brasimone



Unique experimental data



ELTL in Oarai



Bayonet backplate
remote handling
Brasimone



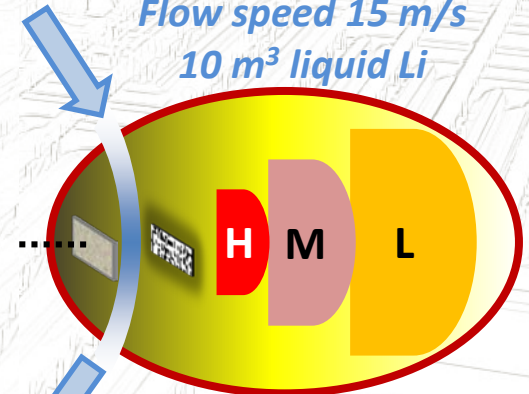
Lithium Target

250 °C

Thickness 25 ± 1 mm

Flow speed 15 m/s

10 m³ liquid Li



World biggest liquid lithium loop

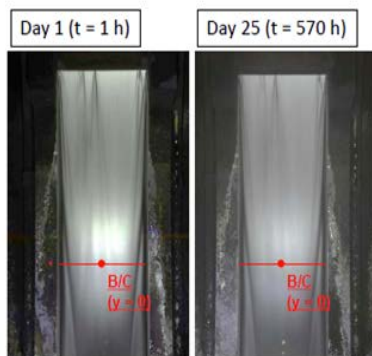
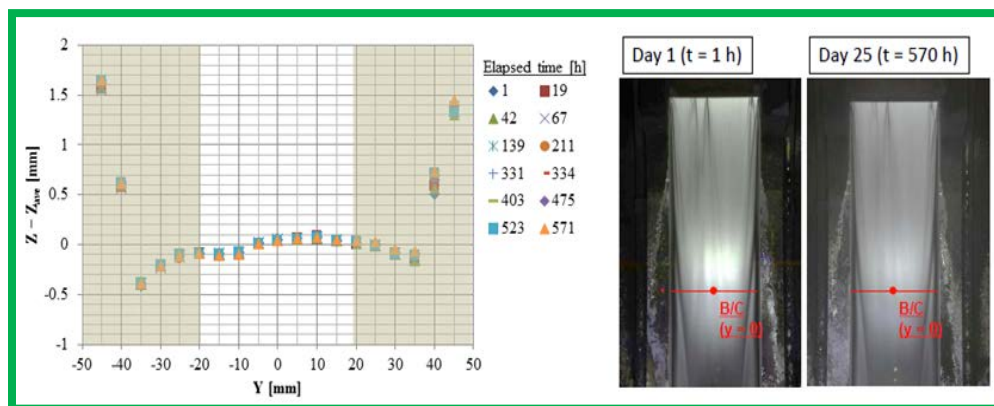
5 m³ of liquid Li



Kondo, H. et al. Validation of IFMIF liquid Li target for IFMIF/EVEDA project, Fusion Eng. Des. 9697, 117122 (2015).

Stable flow of Li target within specifications demonstrated

ELTL



Kondo, H. et al. *Validation of liquid lithium target stability for an intense neutron source*, **Nuclear Fusion** 57 (2017) 066008 (10pp)

Vaporization rates understood

Matching Hertz-Knudsen-Langmuir estimations

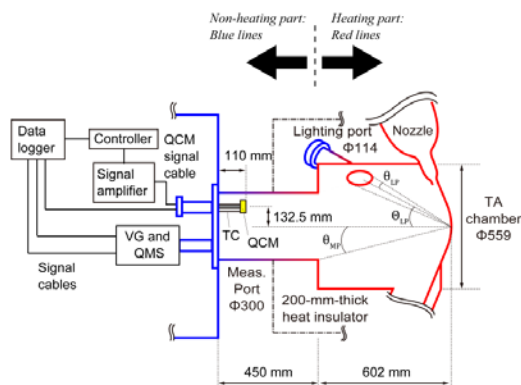


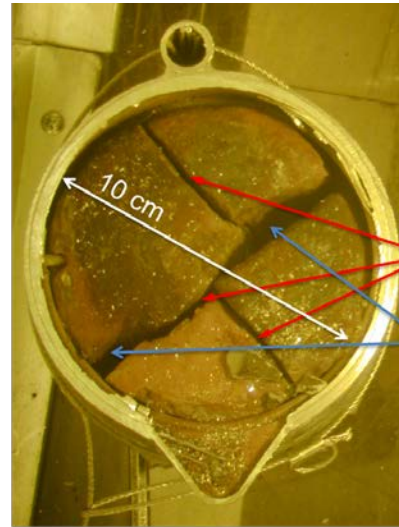
Fig. 4. Experimental setup of the deposition rate and mass measurement. The red and blue parts indicate heating and non-heating components, respectively. The heating components are covered by 200-mm-thick heat insulators.

J. Knaster et al., *An assessment of the evaporation and condensation phenomena of lithium during the operation of a Li(d,xn) fusion relevant neutron source*, **Heliyon** (2016)

T. Kanemura et al., *Analytical and experimental study of rate of evaporation from high-speed liquid lithium jet*, **Fusion Engineering and Design** (2017)

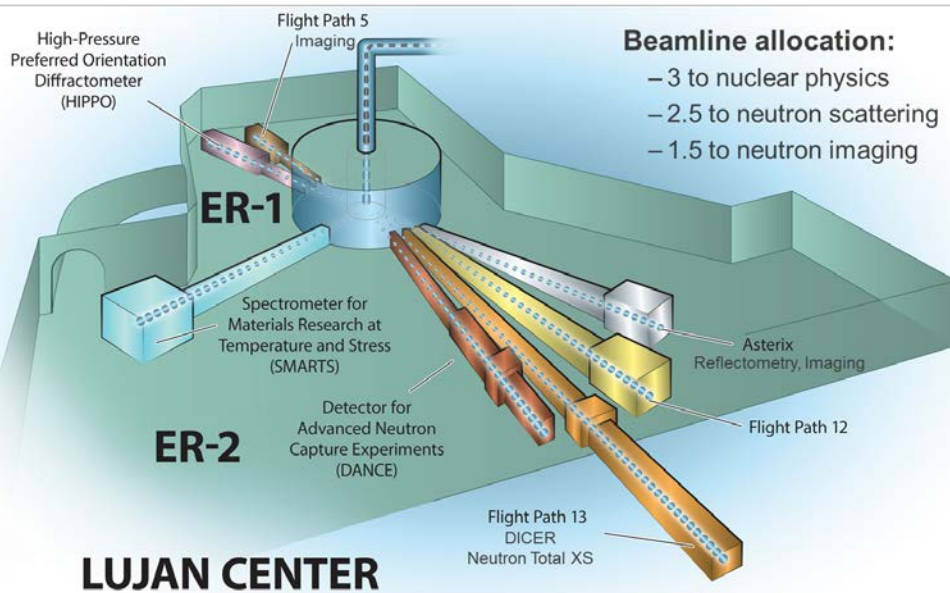
Los Alamos Lujan Center

- 80 kW beam
- Acts as a prototype for higher-power neutron sources



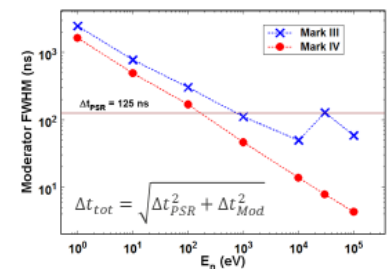
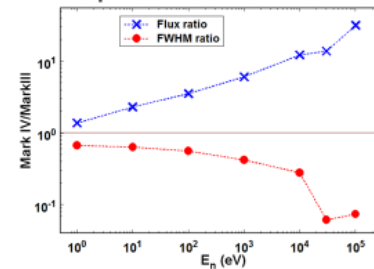
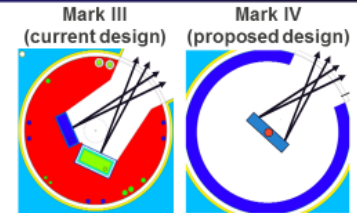
- Target is a solid cylinder of tungsten, edge-cooled
- 80-kW, 800-MeV proton beam
- Top cover plate has been cut off
- Radial cracks from thermal stress
- Portions corroded by highly oxidizing species produced when the proton beam passes through the cooling water

Target from the Manuel Lujan Neutron Scattering Center (LANL)
Source: M.J. Baumgartner, et al., AccApp'03 (2003) 399.



Changes in the Lujan target will improve performance of the nuclear physics beamlines

- Upper tier moderators and inner Be reflector are replaced with a small tungsten target encased in water
- Combined with narrower PSR pulses, large improvements in resolving resonances up to 10 keV are possible



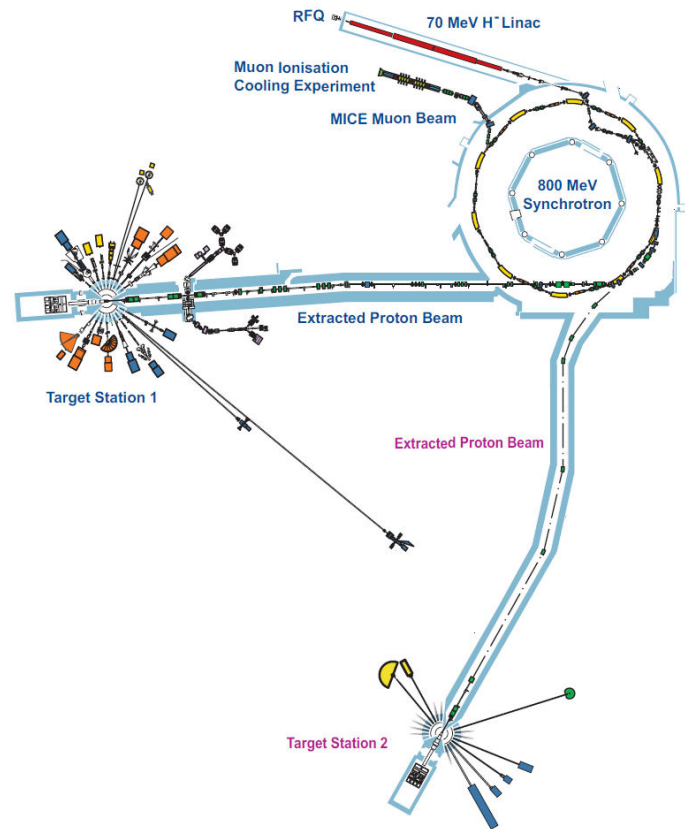
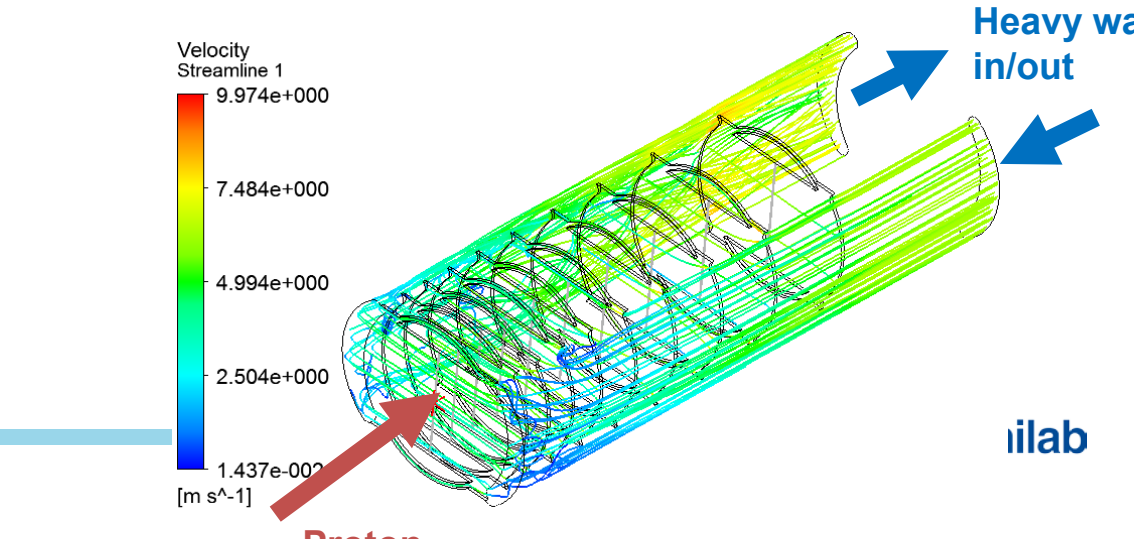
P. Koshier, M. Macko, I. Zavracka, private communication (2017)

STFC RAL ISIS

- 200 kW target, but one of the most efficient neutron sources



Mass of material (kg)	Current TS1	TS1 Project	Reduction
Stainless Steel	73.8	8.1	89.0%
Tantalum	32.7	6.9	79.0%
Tungsten	47.3	46.0	2.8%
Total	153.8	61.0	60.3%





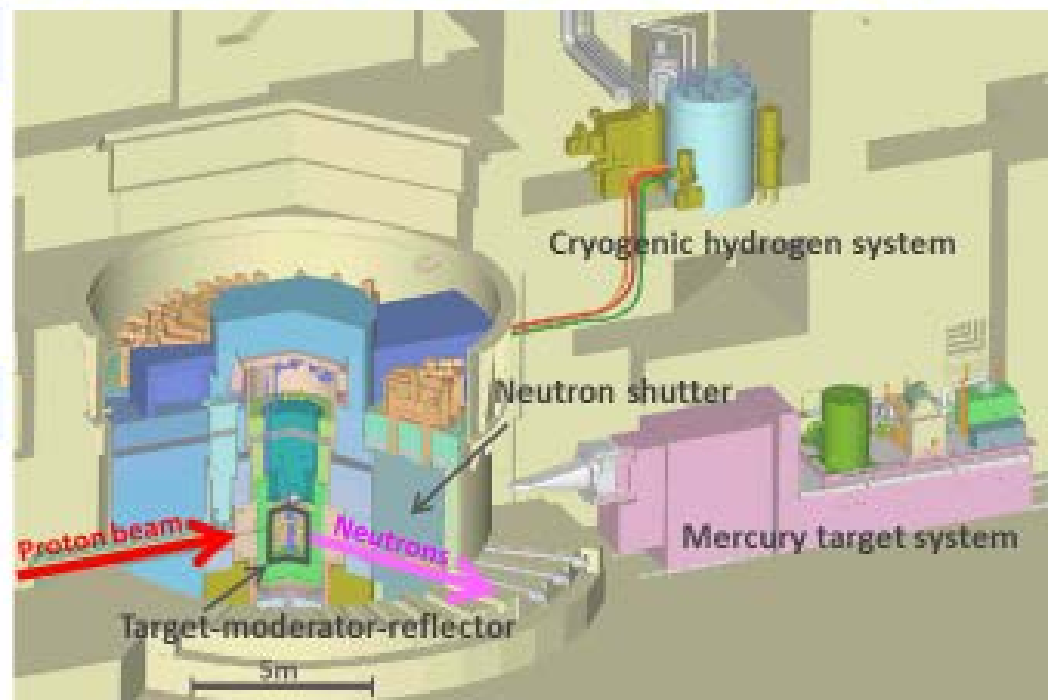
Pulsed spallation neutron source of J-PARC

Proton beam parameters

Proton beam energy	3 GeV
Proton beam current	333 μ A
Beam Repetition rate	25 Hz
Beam pulse length	150 ns
Energy per pulse	40 kJ

Neutron source station

Neutron production target	Hg
Moderator	LH ₂ (para 100%)
Reflector	Be/Fe
Number of neutron beamline	23
Number of neutron shutter	23



Redesigned target vessel #8 has been in use since 2017

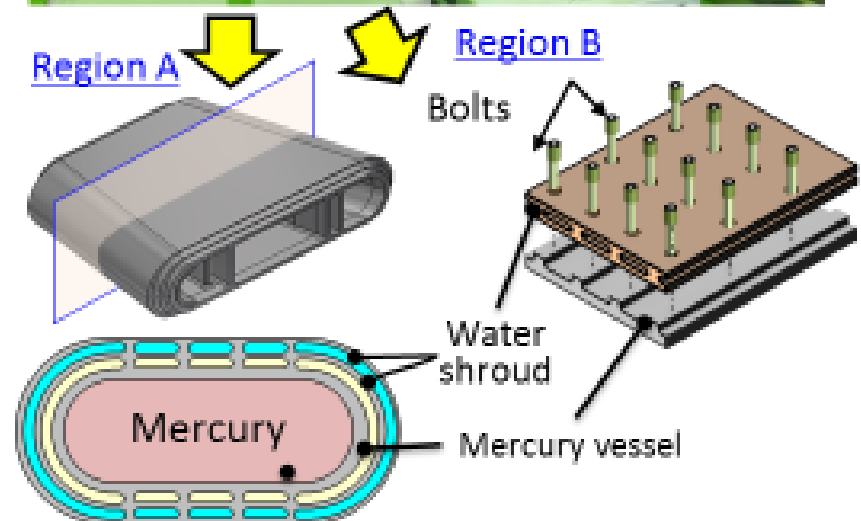
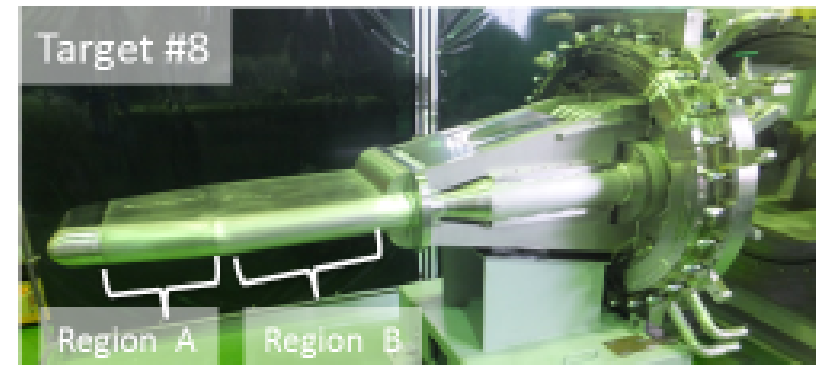
- We have been making efforts to improve structural integrity of the target vessel aiming at achieving stable operation with a power range higher than 500 kW because we had experienced failures in the water shrouds twice at 500-kW operations in 2015.

- **Design philosophy of new target:**

- ✓ Decreasing the risk to make defects at welds which initiates crack propagation
- ✓ Reducing welding lines from the forward part of the target where high stress is induced by temperature swing when incident proton beam is tripped.

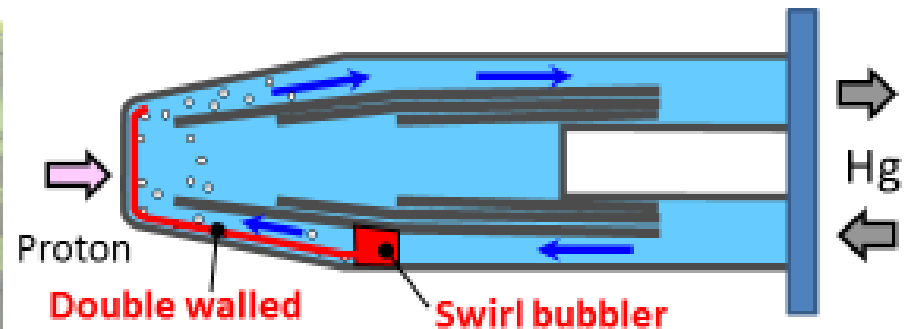
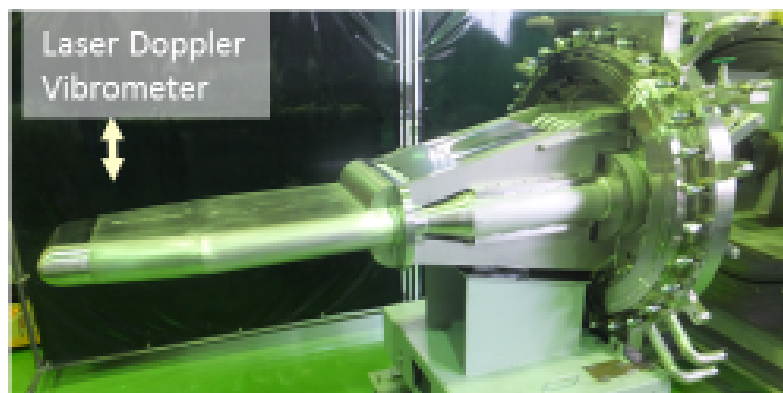


- *New target adopted mechanically coupled structure between mercury vessel and water shroud in the forward part (region A) and that of water shroud in the rear part.*
- *Compared with previous design with bolt joint, length of weld line was drastically reduced 55% in the region A.*



Pitting damage mitigation technique applied to target #8

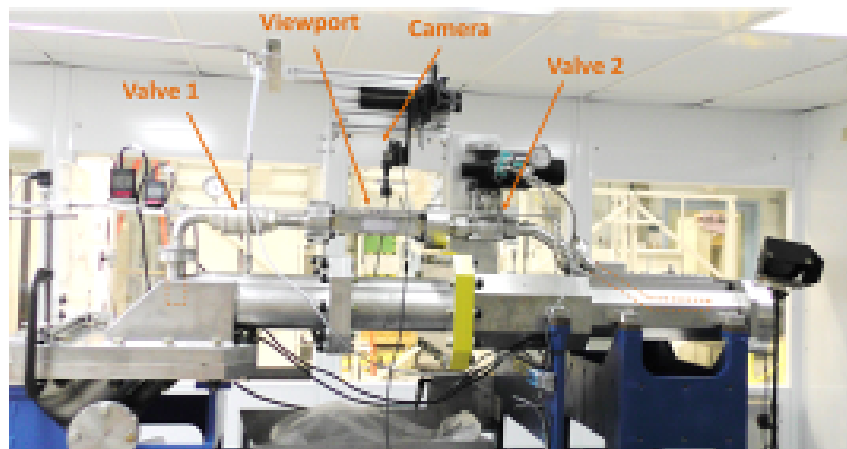
- For the mercury target for short-pulsed spallation neutron source, pitting damage on the front end of the target vessel is a critical issue.
- To mitigate the pitting damage, high-speed mercury flow in narrow channel at the target front end and the gas-micro-bubble injection technique are applied to the redesigned target #8.



- The narrow channel structure has already been in used for the mercury target for the SNS at ORNL.
- Intensities of pressure waves produced in the mercury per proton-beam pulse are 2.4 times higher than those in the SNS target because the repetition rate of 25Hz is lower than 60 Hz at SNS.

Swirl bubbler being tested at the Target Test Facility (TTF)

- Using the Target Test Facility (TTF), bubble sizes are being measured for a swirl bubbler
- Gas hold-up in the mercury loop with swirl bubbler at various flow rates is also being assessed
- Plans are to switch over from the orifice bubbler design to the swirl bubbler in the next design iteration



Target Test Facility (TTF) at ORNL: prototypical target with special features to measure bubble sizes.

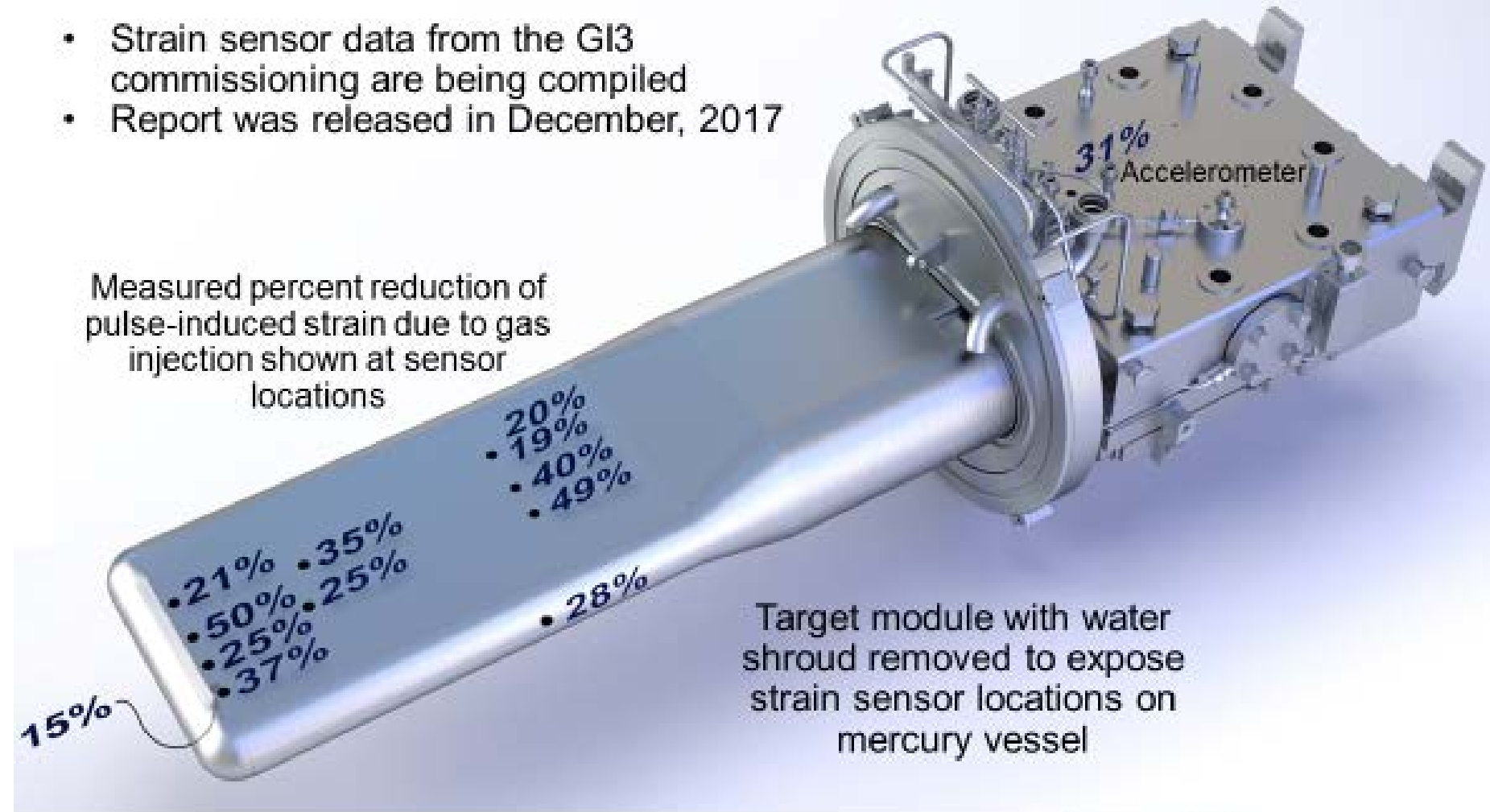


Swirl bubbler

Gas injection mitigates strain due to proton beam pressure pulses

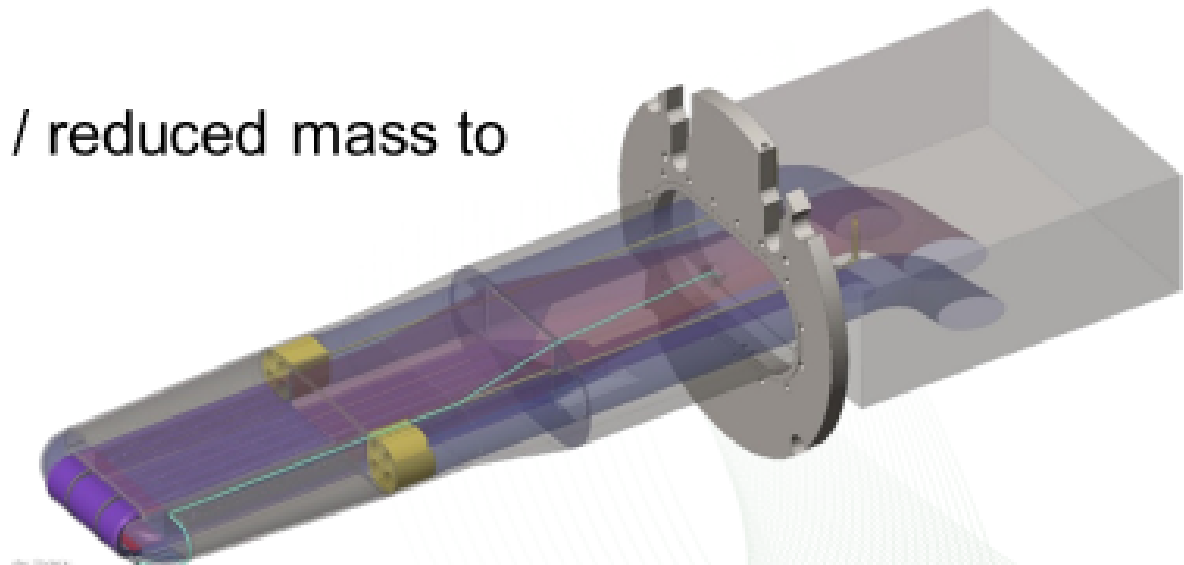
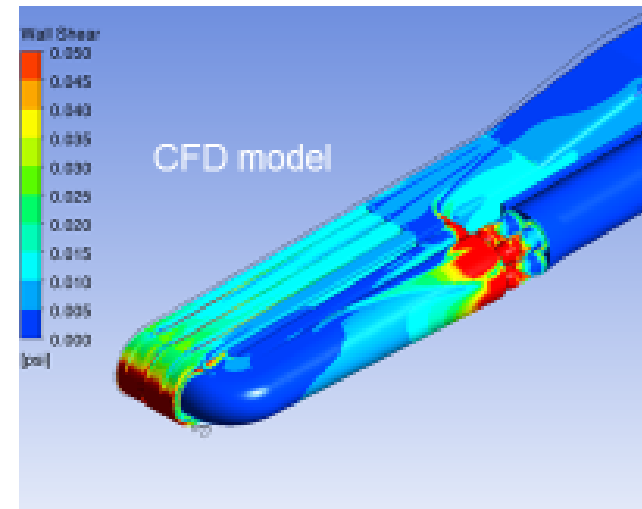
- Strain sensor data from the GI3 commissioning are being compiled
- Report was released in December, 2017

Measured percent reduction of pulse-induced strain due to gas injection shown at sensor locations

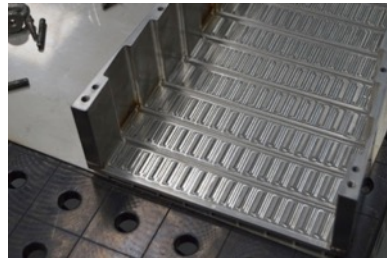
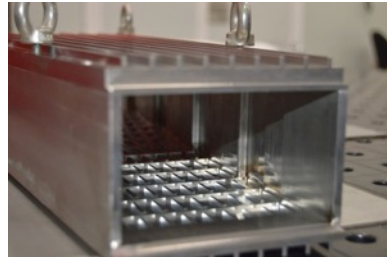
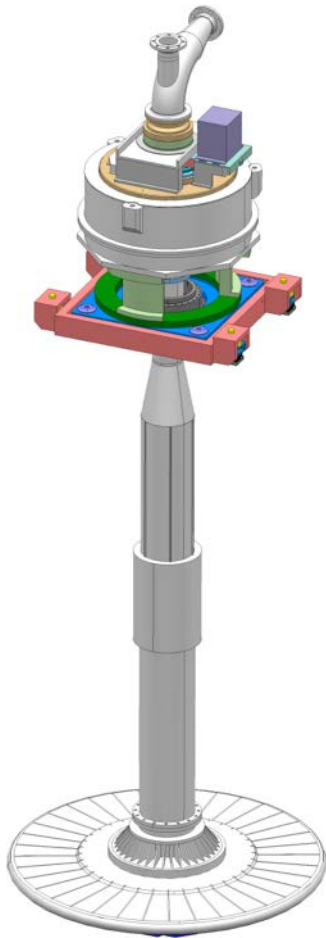


PPU Target (2 MW) conceptual design features

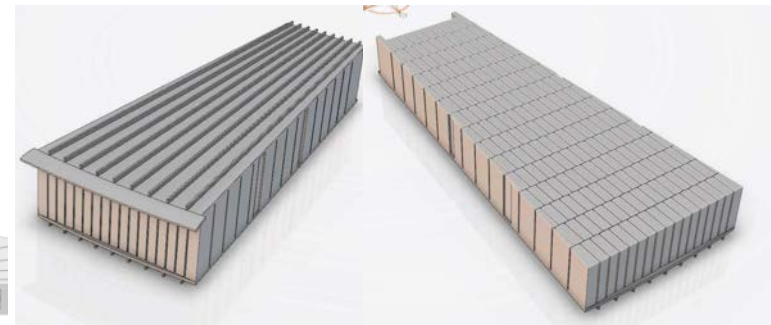
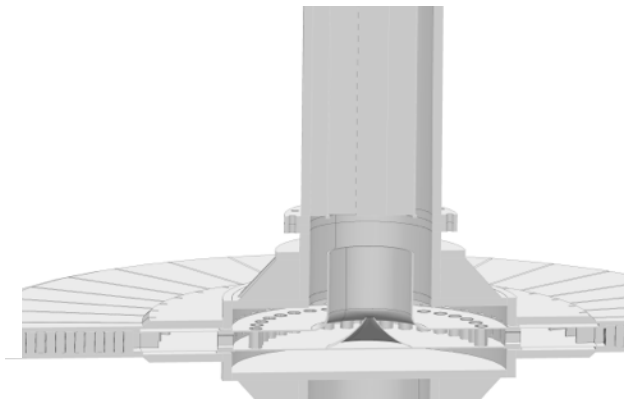
- High-flow small bubble injection with swirl bubblers
- Gas wall at center of inner window
- Tapered front body for directed flow in corners
- Thickened outer beam window in corners
- Stiffer geometry / reduced mass to pulse mode



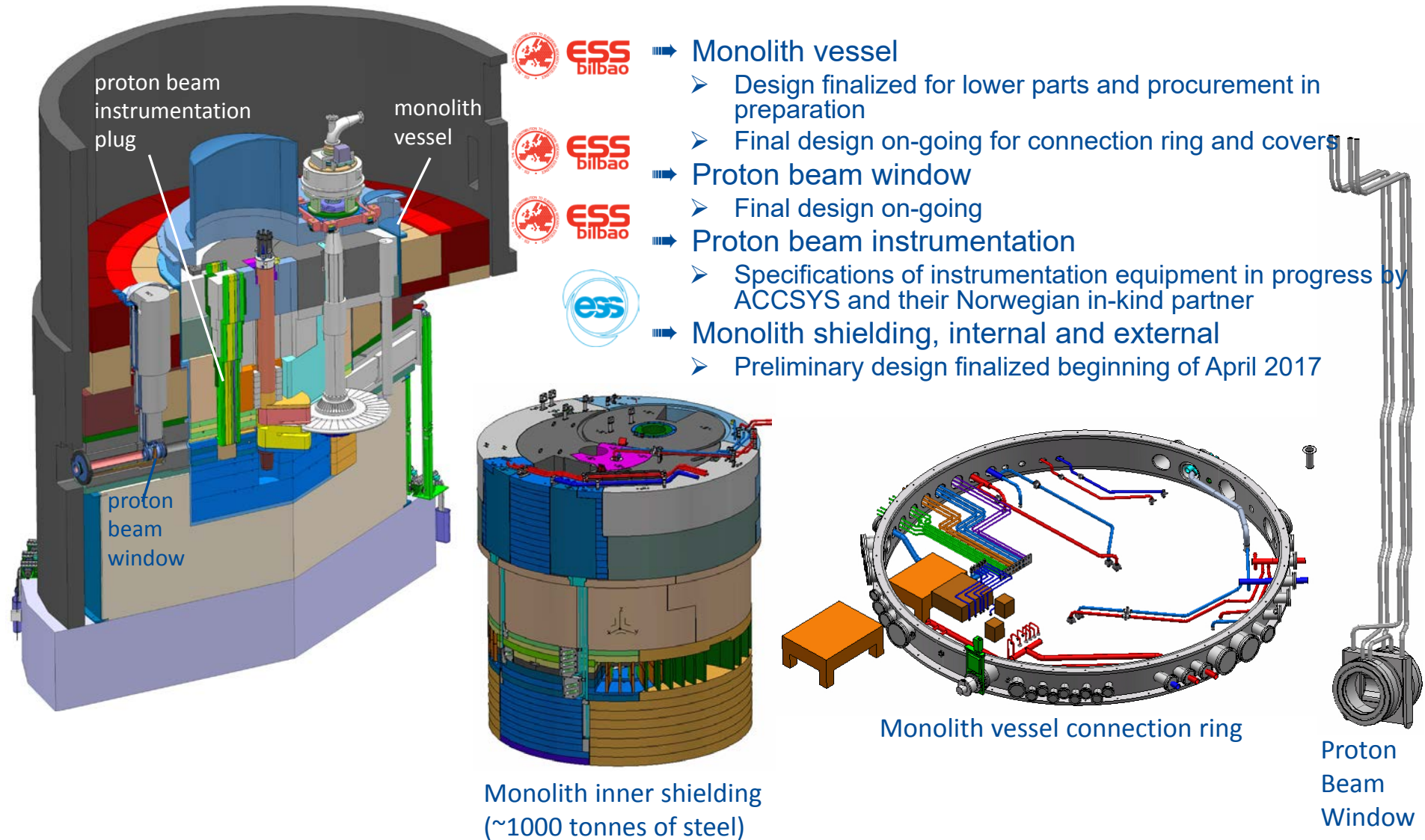
ESS Target: helium Cooled Rotating Tungsten Target



- Tungsten blocks in 36 segments
 - Max 0.1 GW/m³ in tungsten
 - Beam stopping in a meter of tungsten
- Helium coolant
 - Mass flow 3.0 kg/s
 - Pressure 1.0 MPa
- Rotational speed 23.3 rpm
- Wheel diameter 2.6 m
- Shaft length > 5 m



ESS Monolith systems



ESS Target Systems



ESS
bilbao

- Target wheel and drive unit
 - Design of cassettes and spallation material is finalized and procurement is imminent
 - Manufacturing of prototype is on-going
 - CDR for Target Wheel and Shaft planned in May 2017

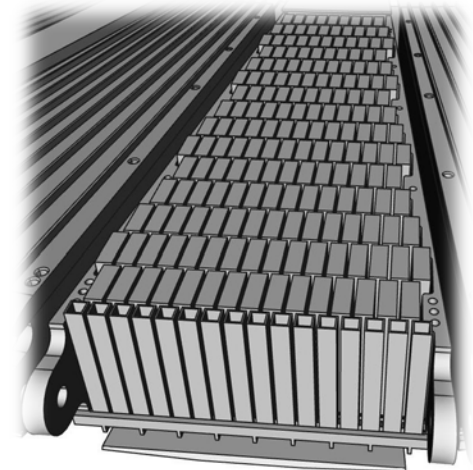


Progress on Target Helium Cooling System

- Orders placed for helium circulators
- Detailed design is on-going for the rest of the cooling loop

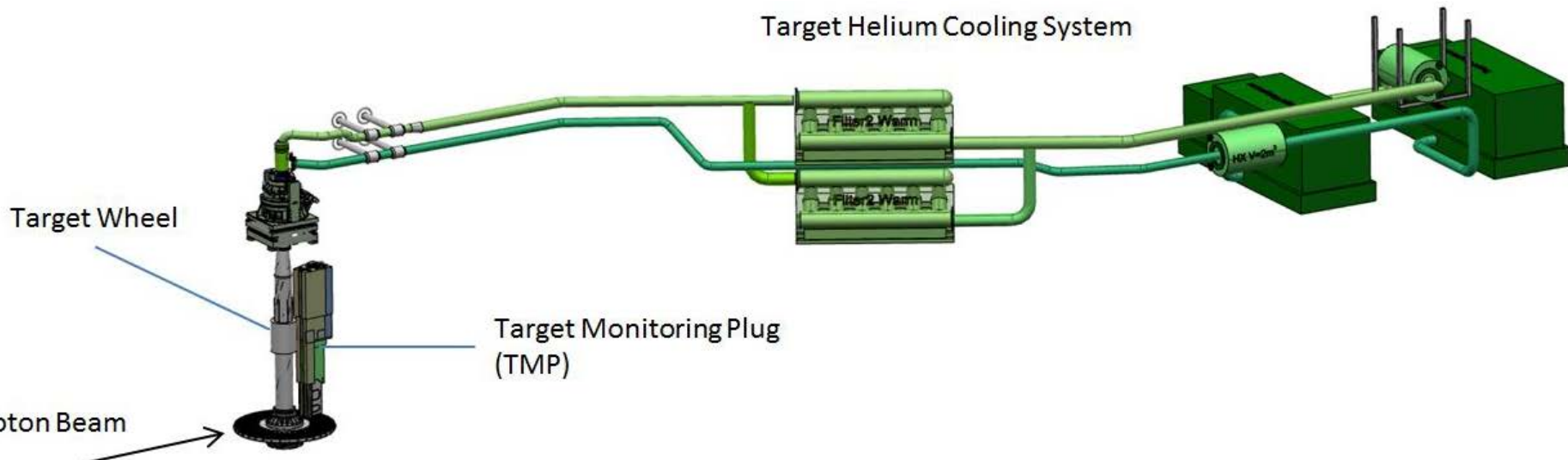


- Target monitoring
 - Conceptual solutions defined and preliminary design about to start



Tungsten Bricks in Cassette

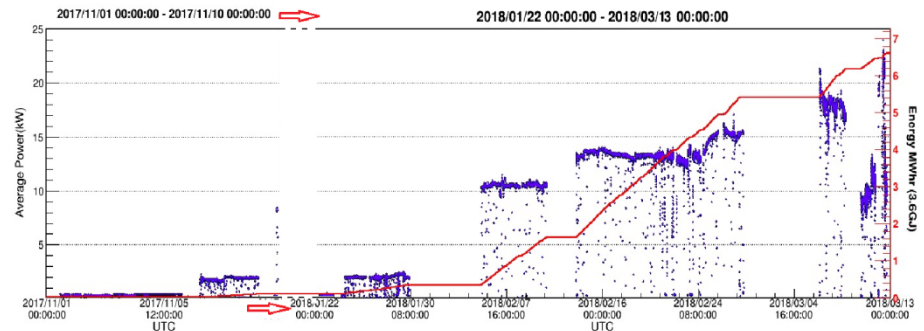
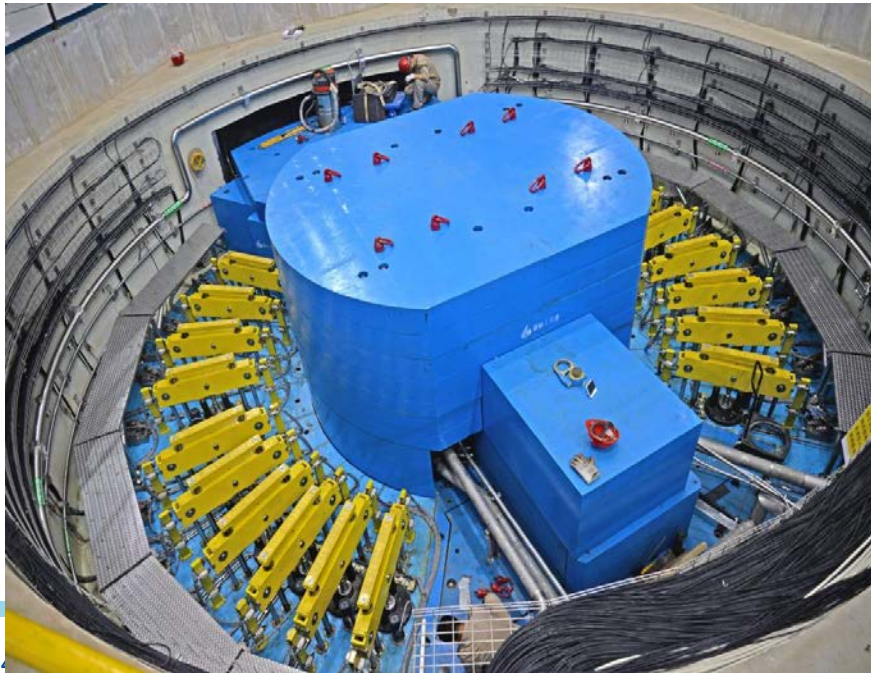
Target Helium Cooling System



CSNS Target Station

Phase-I: 100 kW, Upgrade plan: 500 kW

- Aug.28,2017, first beam on Target, March.24, 2018, first publication on Li-Battery research, present operate at 25-30kW , 50kW by the end of 2018.
- Challenges of high-power stations: high heat deposition on target materials and moderators, high decay heat
- R&D activities : new W-Ta cladding technique, Radiation damage on Ta-W interface, target imaging system, Rotating target concept etc.

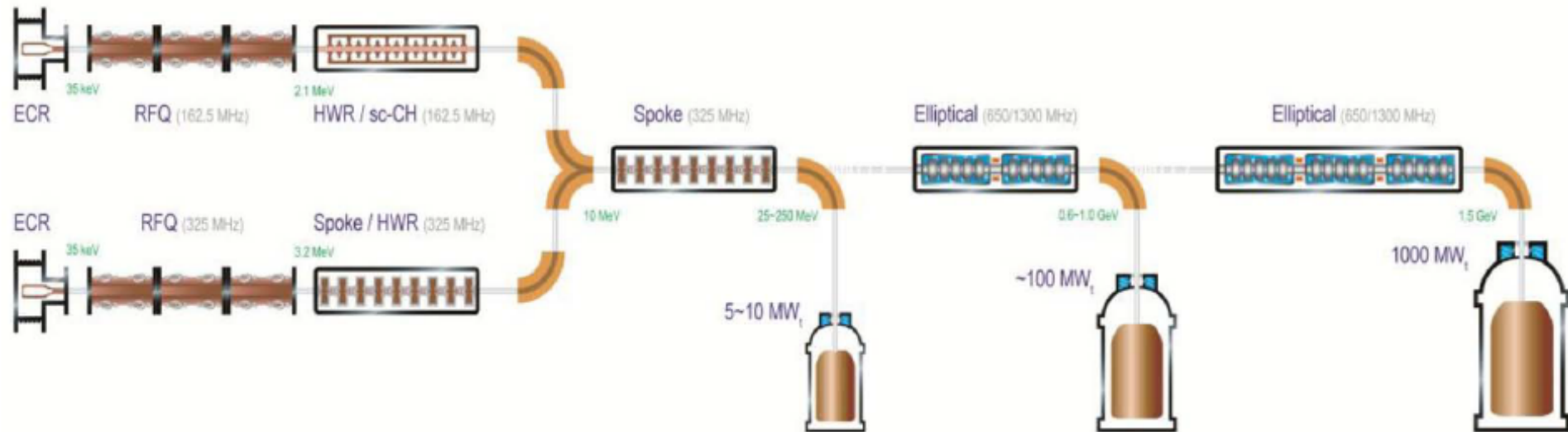


Chinese ADS roadmap

2010

2016

2022



CIADS: INITIAL FACILITY

250MeV@10mA

5-10MW

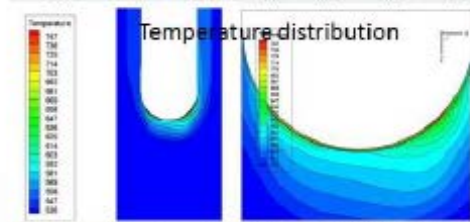
DEMO FACILITY

INDUSTRY FACILITY

Design goals: For CIADS, the target can accommodate 2.5MW=250MeV@10mA; the target system can update to ~50 MW=2.5GeV@20mA.

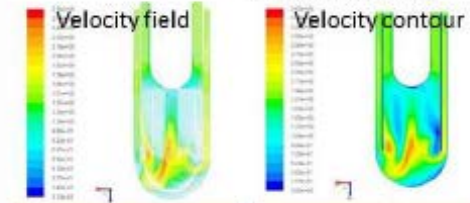
LM Window target research

T91 parameters						
Parameter	100	200	300	400	500	600
Heat transfer coefficient (W/m²K)	10.0	10.0	11.0	10.0	10.0	10.0
Thermal conductivity (W/mK)	16.0	16.0	16.0	16.0	16.0	16.0
Beam size (mm)	100	100	100	100	100	100
Beam power (MW)	10	20	30	40	50	60
Beam spot size (mm)	10	10	10	10	10	10
Beam position (mm)	100	100	100	100	100	100



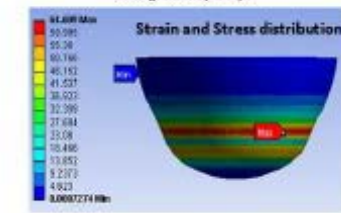
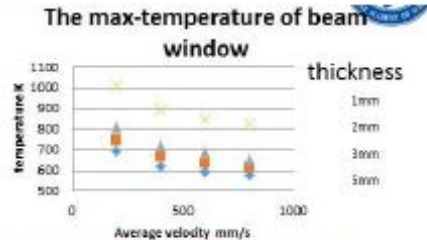
Average velocity: 0.25m/s

Temperature different between the inner surface and out surface of window is ~50K.

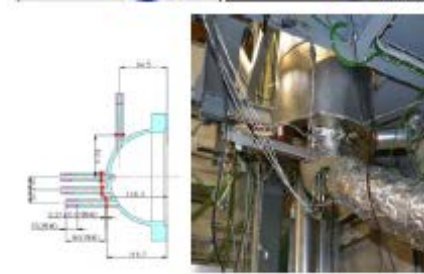
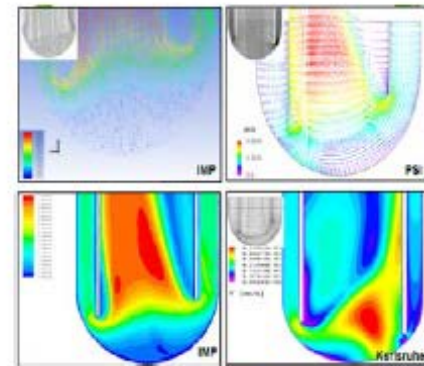


Average velocity: 0.76m/s

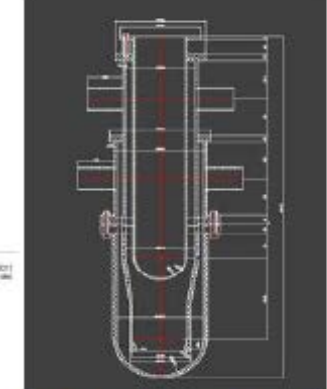
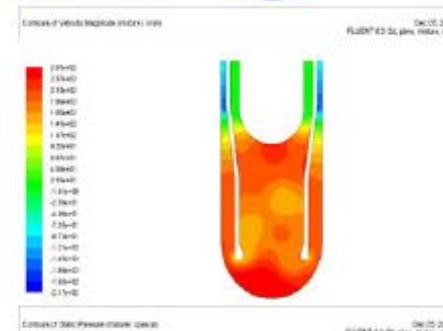
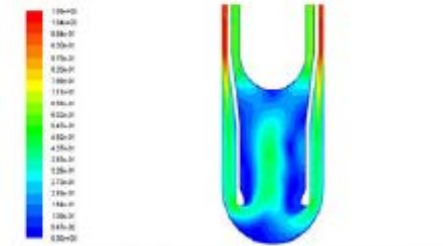
Average velocity: 0.76m/s



Hydrodynamic-Structural-Heat Analysis: parameters	
structural safety factor	~ 3
Thickness of window	2.0 ± 0.5mm
Inlet temperature	525 ± 25 K
Average velocity	0.3 ± 0.05m/s



- Average Beam intensity : $\sim 30 \mu A/cm^2$. For an ADS, If the diameter of the beam pipe can choose $\sim 30-40$ cm, then, the target would be design for $>10MW$.
- The heat removal of the window will be limited by the heat conduction of the target material and convection-cooling; the life time of the window will be limited by the radiation damage and al.

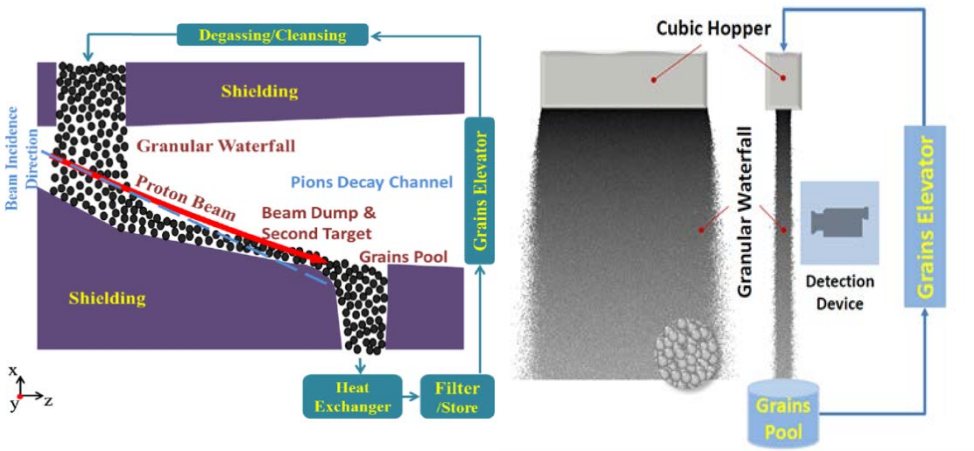
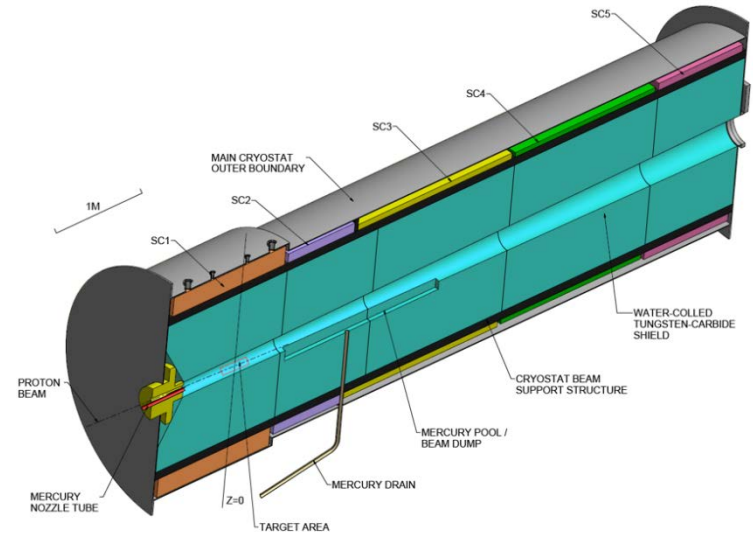


MOMENT Target Station

MOMENT: ADS-type proton driver, 1.5 GeV- 15 MW, for producing muon-decay neutrino beams .

Baseline concept:

- Hg jet target following Neutrino Factory
- SC capture solenoid 14 T
- Spent proton extracted
- Very high but steady heat deposition
- Very high radioactivity, also toxicity

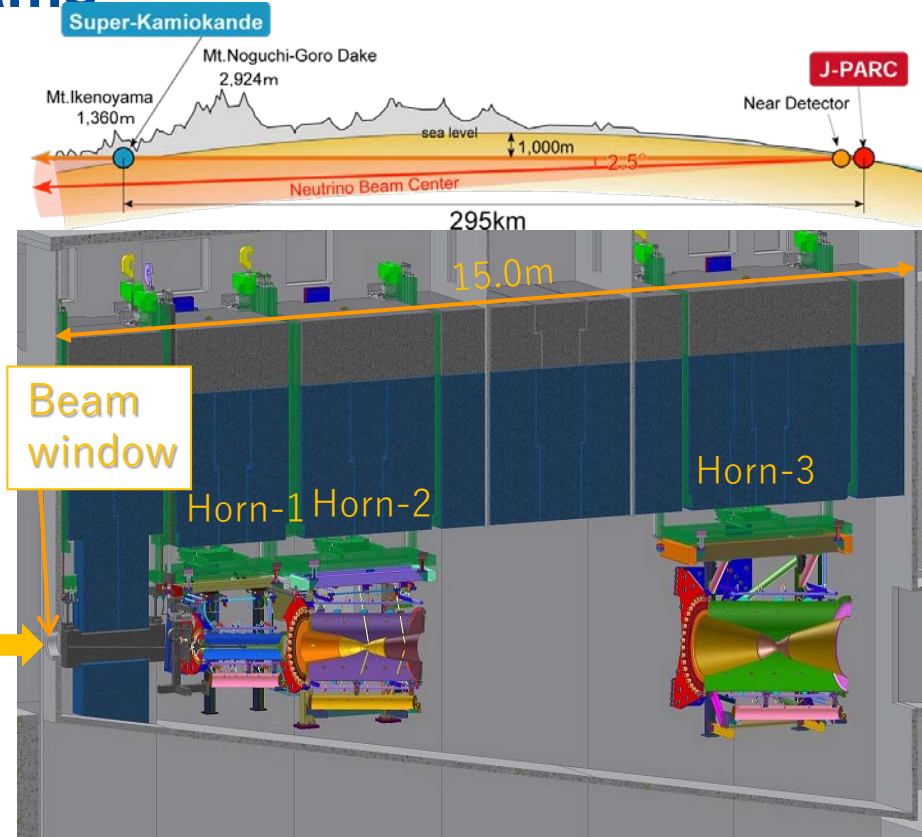
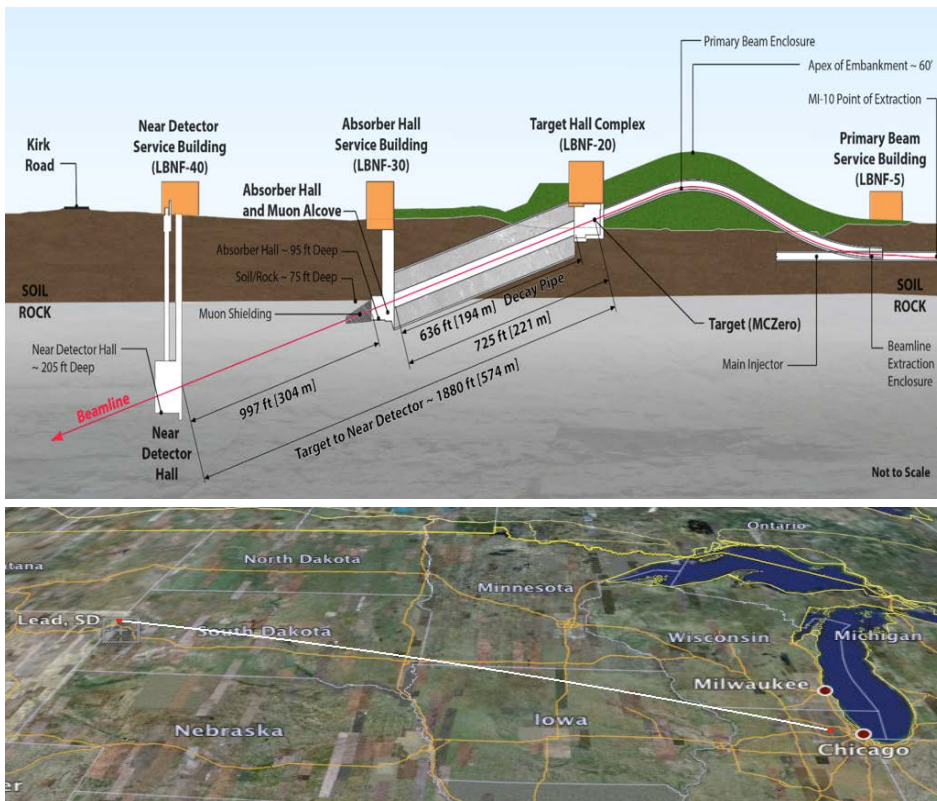


Favourable option:

- ◆ Waterfall-like target of tiny tungsten balls, external heat exchange
- ◆ Simulation shows it is promising, prototype test will be carried on

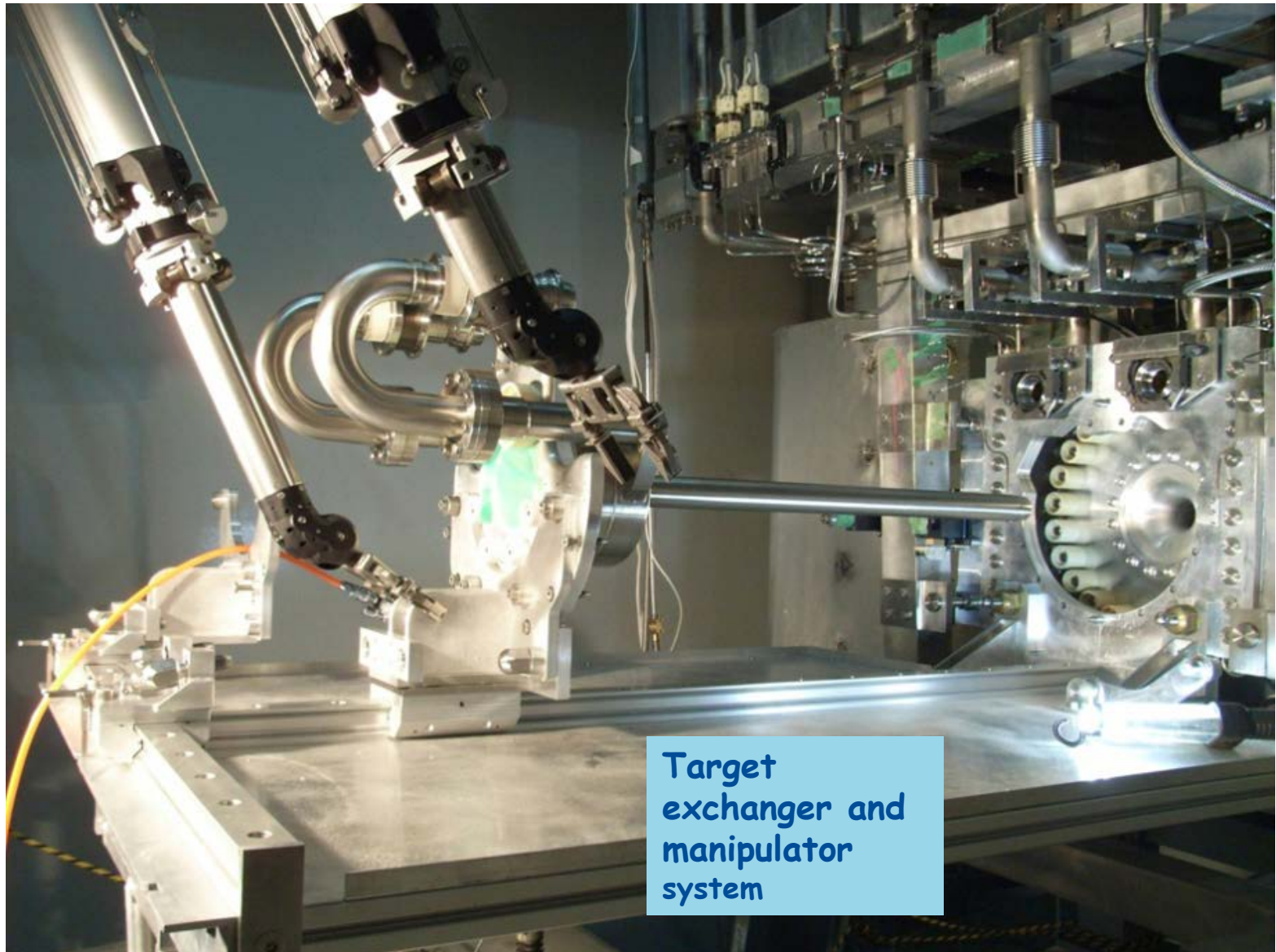
R&D efforts on Muon production and capture, spent proton extraction are undergone at CSNS muon source

New & Upgraded Neutrino beams



- Fermilab & J-PARC beams to long-baseline experiments (300-1300 km)
 - Facilities built to accommodate ~ 4 MW of beam
- Fundamental neutrino and particle physics

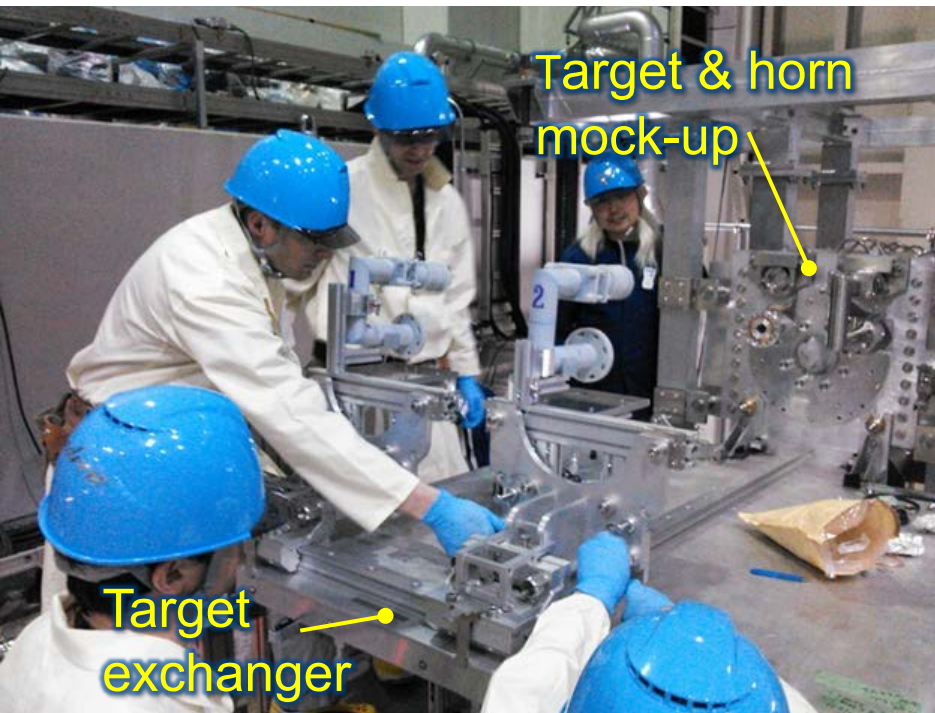
Remote Target Exchange System



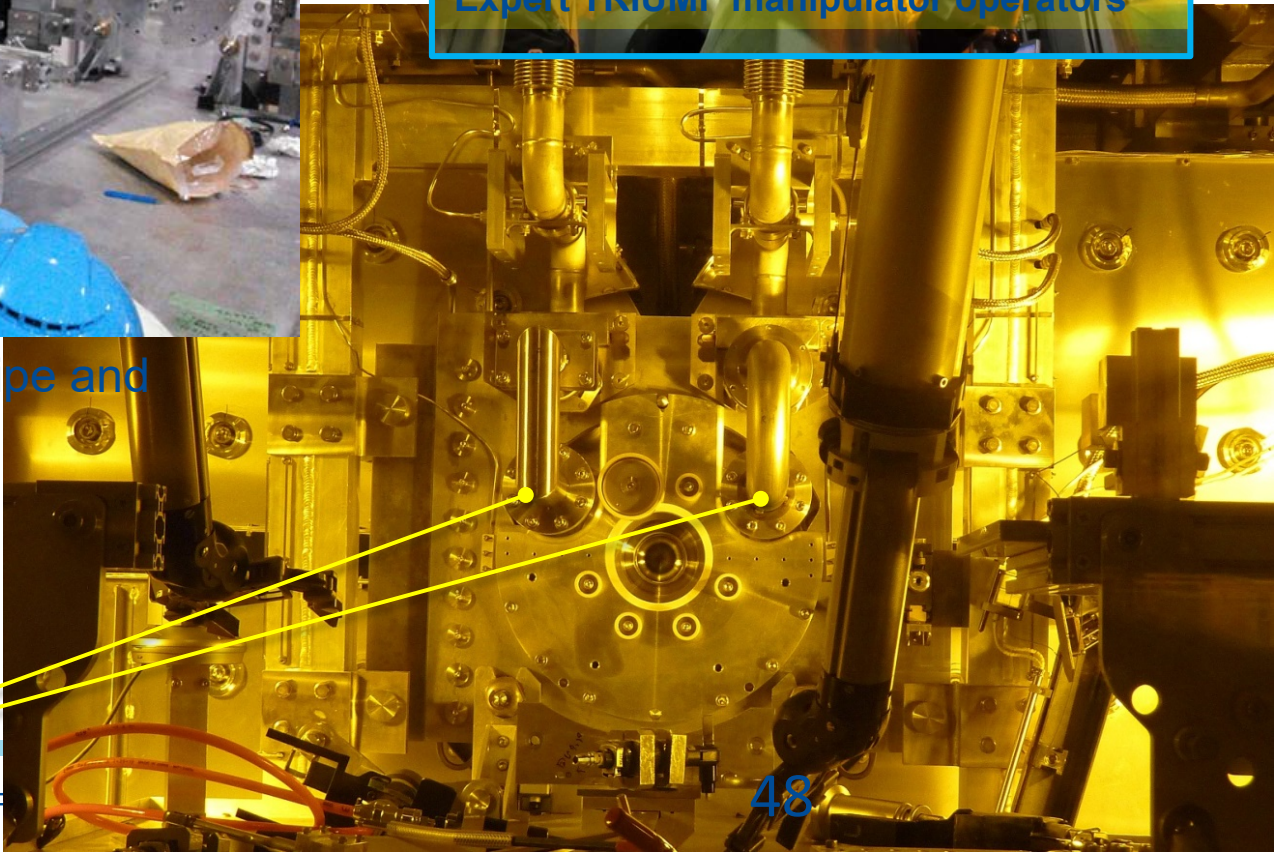
Target
exchanger and
manipulator
system

Helium pipe replacement

2015年12月17日



Rehearsal using 3D printed pipe and modified target exchanger

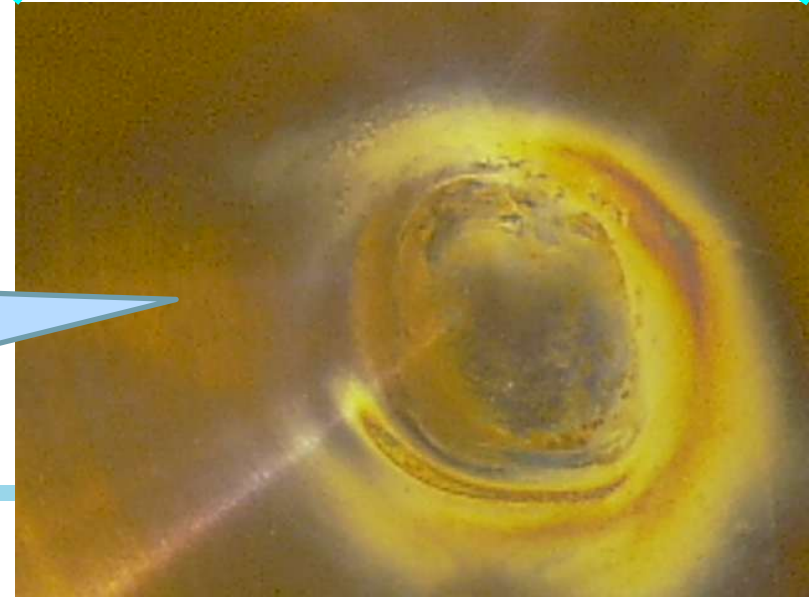
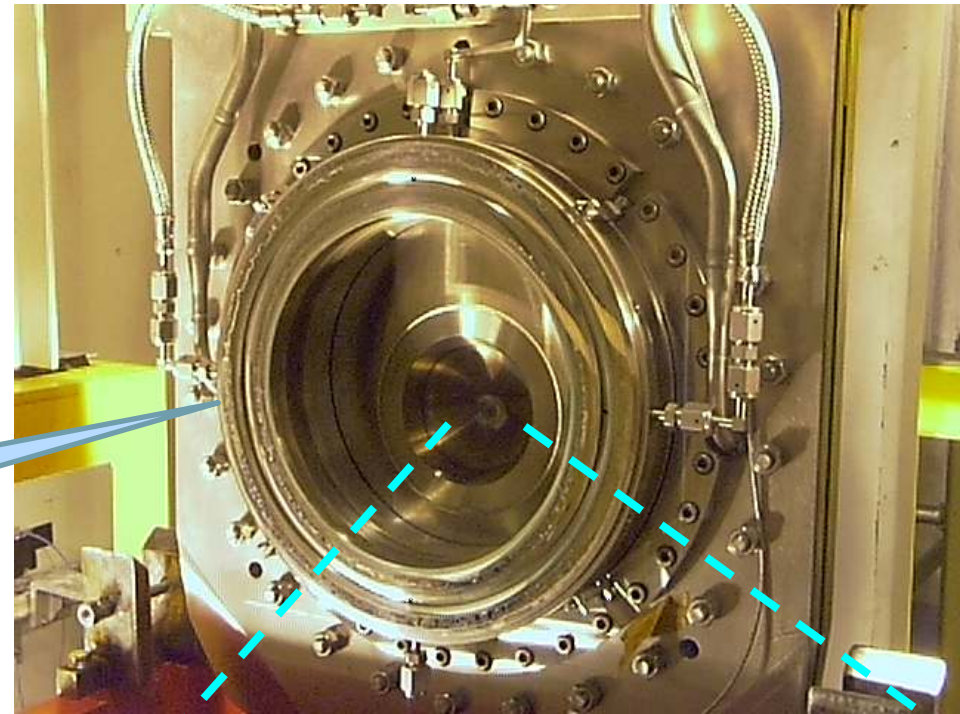


Beam window replacement 22 Aug 2017

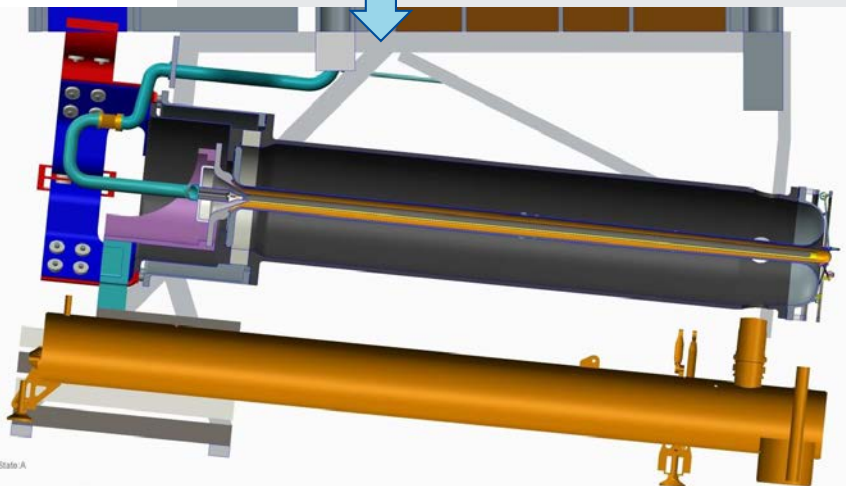
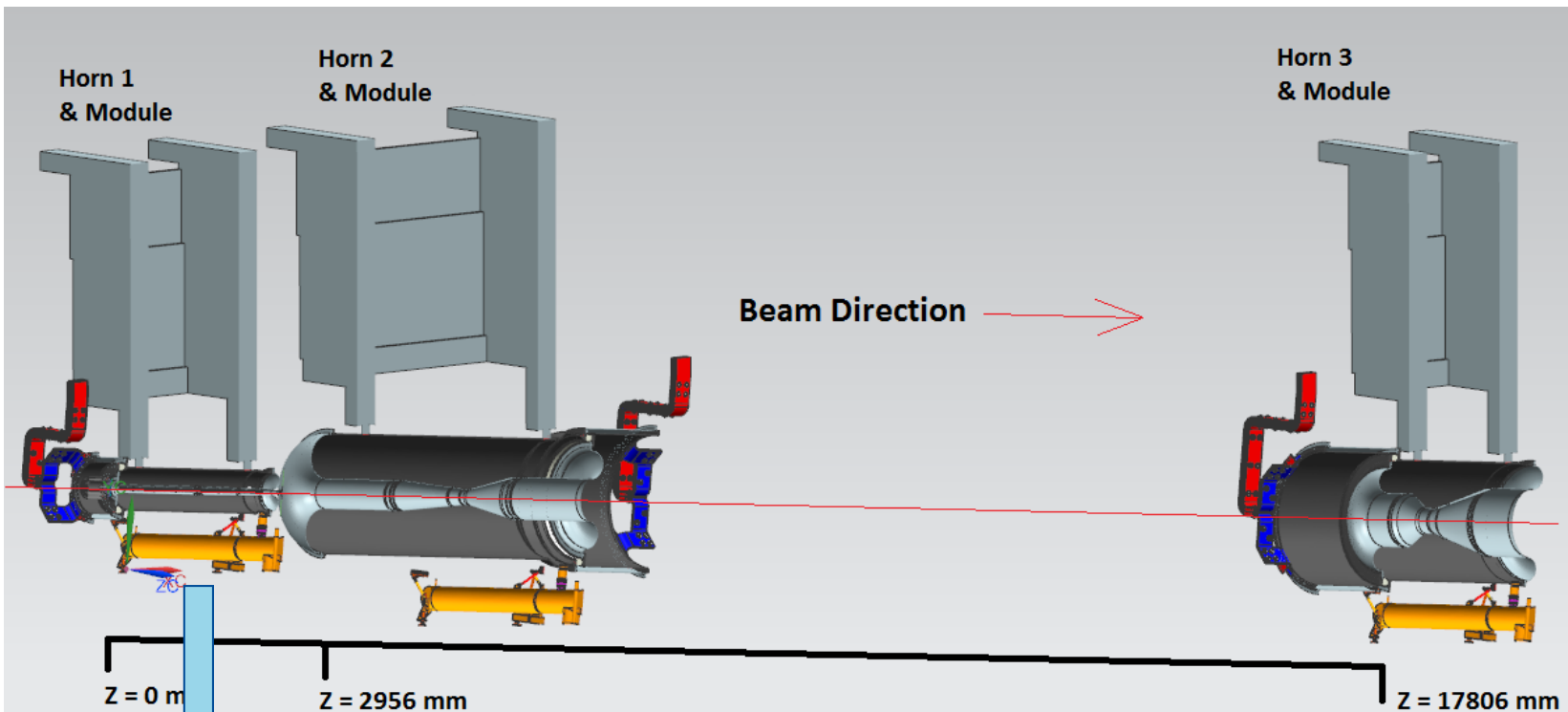
- Want to avoid failure of window during operation
- -> Decided to replace with spare during 2017 scheduled shutdown

- Inflatable pillow vacuum seal appears deformed after removal
- Not possible to reuse the old beam window.

- Centre of beam window appears to be damaged / discoloured
- Need to conduct Post-Irradiation Examination -> talking to materials scientists

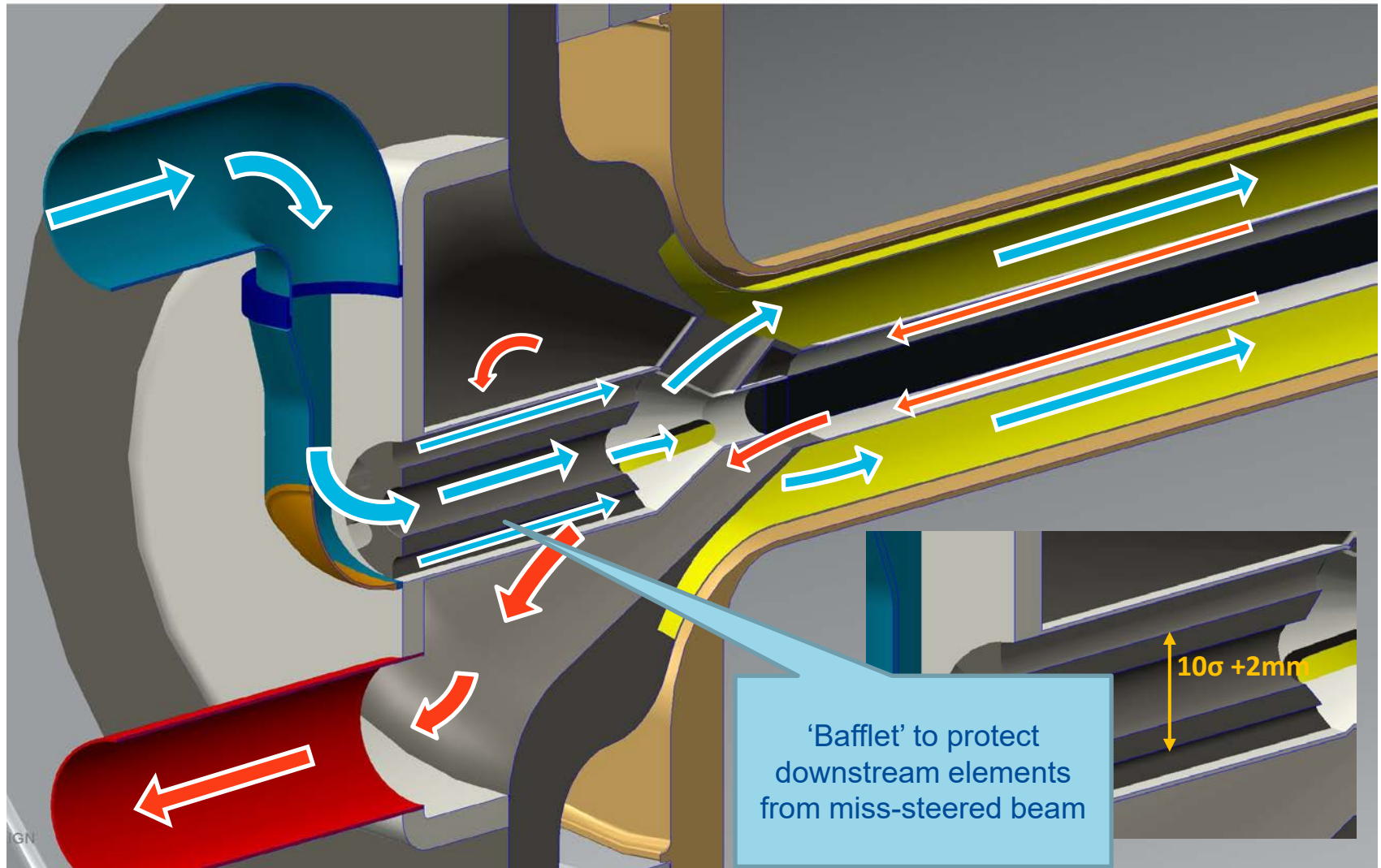


LBNF Target & horns optimized by genetic algorithm

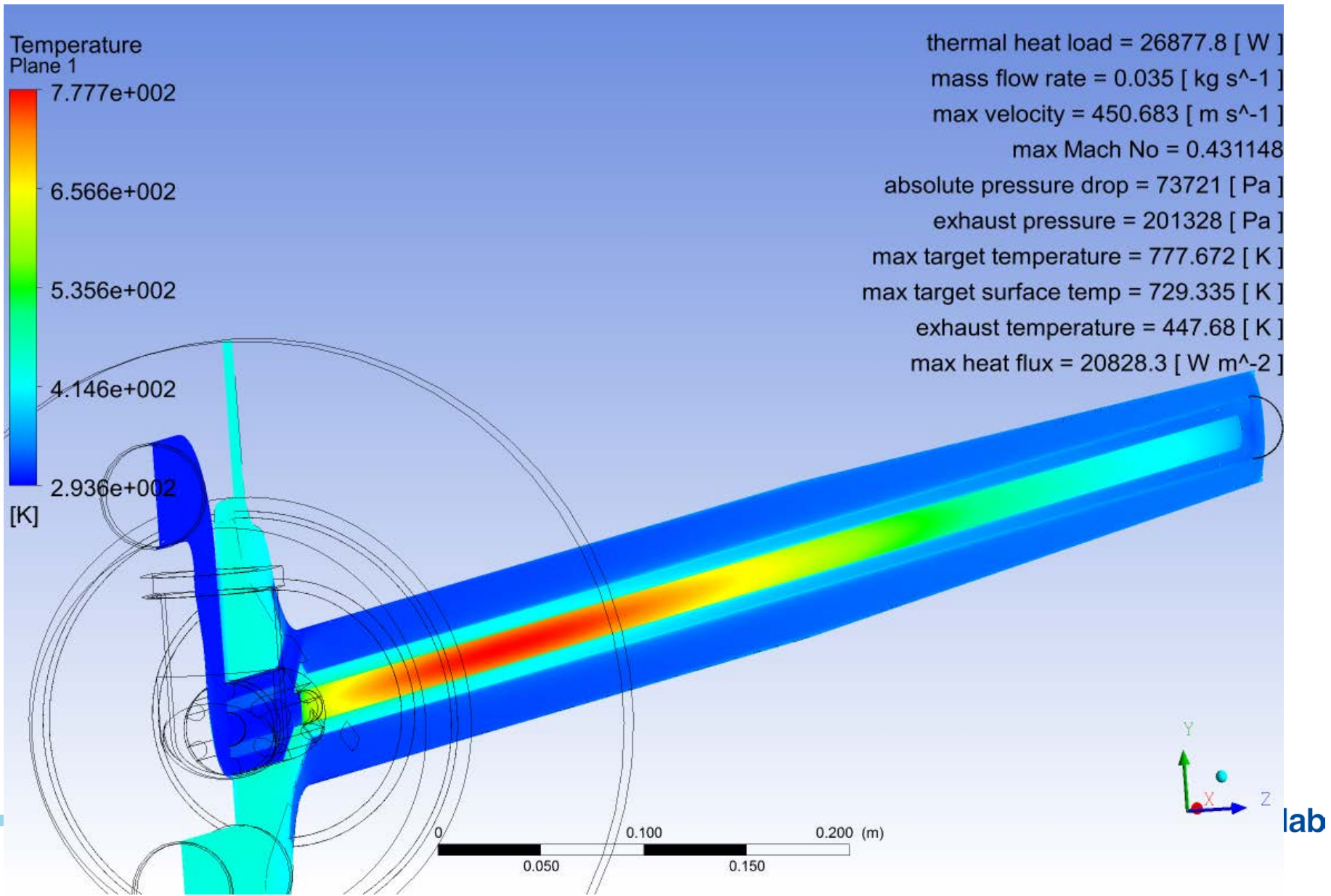


- 3 horn system (like T2K)
- New target plan:
- Change from water (like NuMI) to helium cooled graphite (like T2K)
 - Long ($4\lambda_{\text{int}}$ 2 m): higher yield
 - fewer on-axis wrong-sign pions
 - but needs downstream support

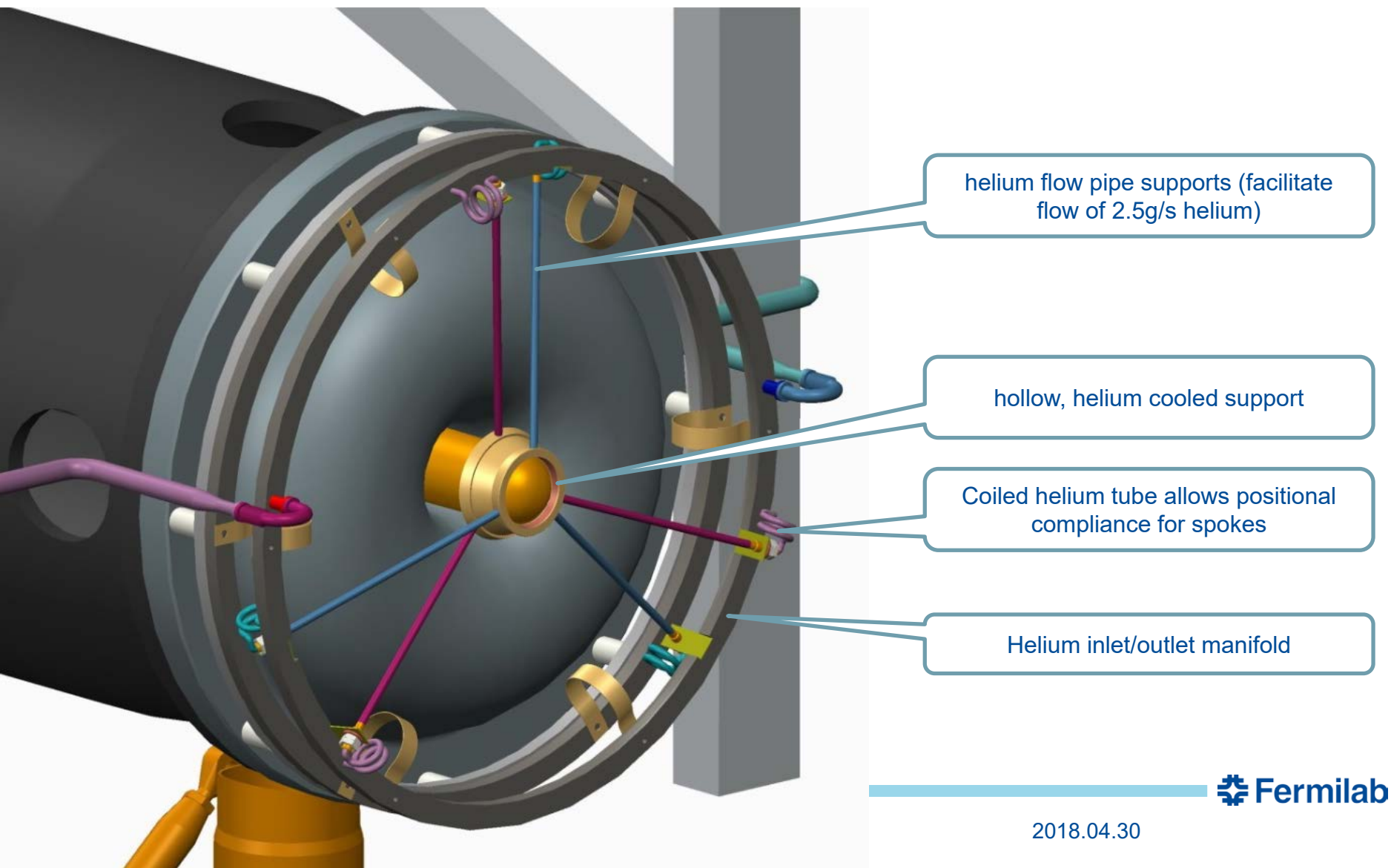
Optimised helium cooled target proposal



Temperatures at 1.2MW steady state simulation



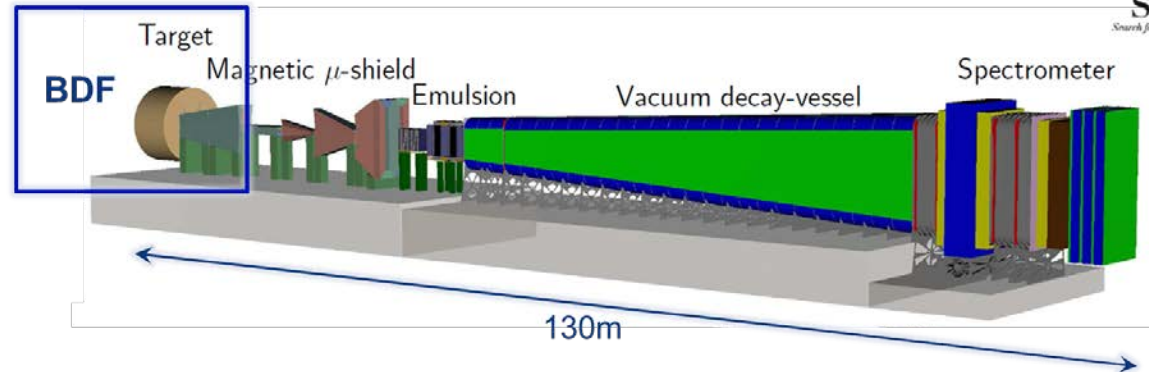
2 m long target: needs actively cooled downstream support



CERN Beam Dump Facility

- Absorb full-power of SPS beam
 - Up to 400 kW

SHiP Comprehensive Design Phase



1st part: TZM core

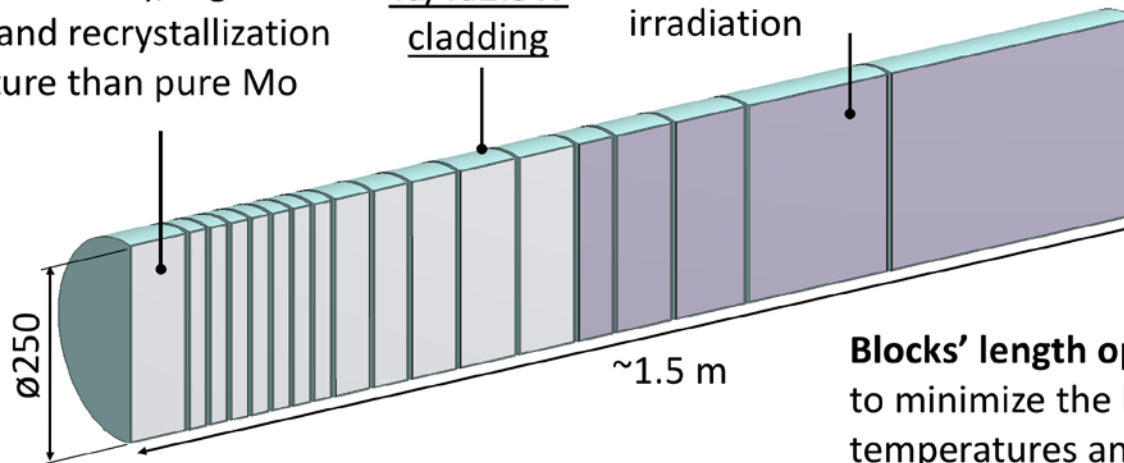
Molybdenum alloy, higher strength and recrystallization temperature than pure Mo

Ta/Ta2.5W cladding

2nd part: Tungsten core

High-Z and good performance under irradiation

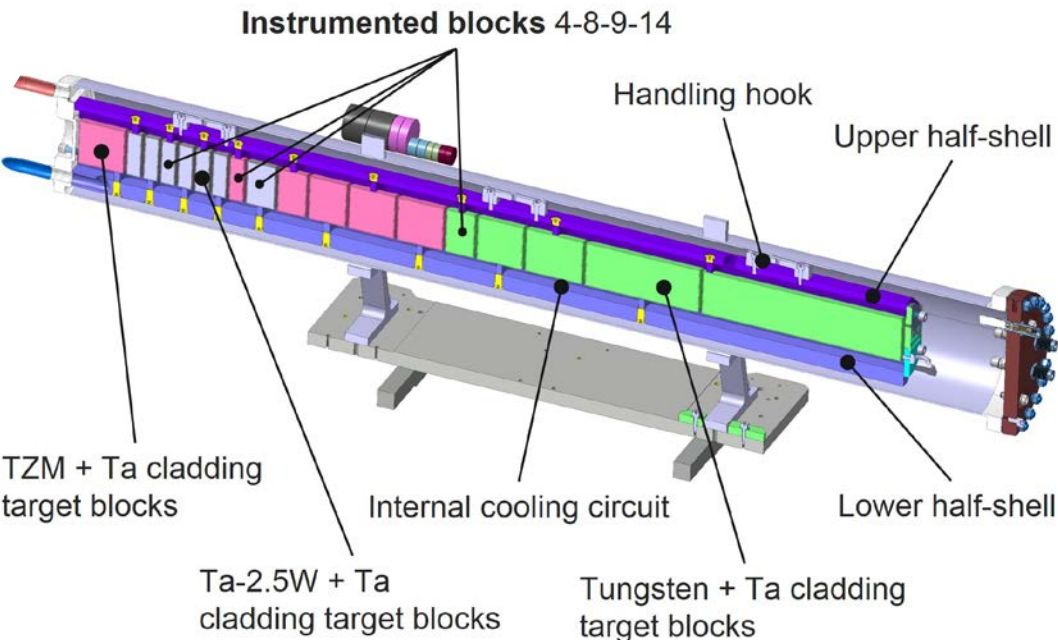
5 mm gap between blocks



Blocks' length optimized to minimize the level of temperatures and stresses

BDF target prototype beam test

- Beam test end of 2018 in the North Area (20 kW power deposited)
 - **Reproduce the level of temperatures and stresses of the final target** → 1-s slow beam extraction + non-diluted beam + $3\text{-}4 \cdot 10^{12}$ p+/cycle
 - **Crosscheck the FEM calculations performed** → several instrumented blocks: strain-gauges + optical fibers + PT100
 - **Post Irradiation Examination** after irradiation → remote opening and extraction of the blocks



Outline & Summary

- Targets have a long history, but are entering a new regime with multi-MW beams (mostly protons)
 - Numerous challenges will confront the efficient use of high-power beams
 - Generic R&D can make an impact
 - Understanding of materials
 - Development of technologies
- RaDIATE Collaboration formed to coordinate research
radiate.fnal.gov
- Facilities world-wide are under active development and performing specific R&D and design

High-Power Targetry Workshop 2018, June 4-8, Michigan State University

<https://indico.fnal.gov/event/15204/>

Acknowledgements

- Marco Calviani (CERN)
- Chris Densham (STFC RAL for T2K / LBNF)
- Patrick Hurh (Fermilab for LBNF / R&D)
- Taku Ishida (KEK/J-PARC for T2K)
- Juan Knaster (IFMIF EVADA – now ITER)
- Yong Joong Lee (ESS)
- Frederique Pellemoine (FRIB)
- Eric Pitcher (LANL PSR)
- Jingyu Tang (IHEP for CSNS / MOMENT)
- Mark Wendell (ORNL SNS)
- Dan Wilcox (STFC RAL for ISIS)



Multi-MW Targets for Next-Generation Accelerators

Robert Zwaska,

The 9th International Particle Accelerator Conference (IPAC '18)

30 April 2018